



OPEN An informed monotone neural network for multi-stage chronic kidney disease prediction with integrated GFR estimation

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Chronic kidney disease (CKD) is a progressive condition affecting over 850 million people worldwide, where timely detection and accurate staging are critical to reducing morbidity, mortality, and healthcare burden. Current machine learning approaches often treat CKD prediction as a binary task, neglecting the ordered nature of disease stages, and rarely incorporate physiological constraints or calibrated probability estimates essential for clinical decision support. We propose a multi-stage CKD prediction framework that integrates ordinal classification, probability calibration, and a serum creatinine-monotonicity constraint within a knowledge distillation paradigm. A calibrated CatBoost model serves as a teacher, transferring temperature-scaled class probabilities to an ordinal neural network student augmented with an auxiliary eGFR regression task. Evaluated on a cohort of 750 patients from Al-Ramadi Teaching Hospital, our approach achieved superior macro-F1 and reduced expected calibration error. This work demonstrates that combining ordinal learning, calibration, and physiological constraints yields models that are not only accurate but also aligned with clinical reasoning, offering a pathway toward safer and more trustworthy AI tools for CKD management. Limitations like circular reasoning and target leakage are also discussed.

Chronic kidney disease (CKD) is a progressive, often silent condition affecting more than 850 million people worldwide, making it one of the leading contributors to global morbidity and mortality¹. Driven by ageing populations and the rising prevalence of diabetes and hypertension, CKD incidence continues to increase, exerting a substantial burden on patients, healthcare systems, and economies. In early stages, CKD is largely asymptomatic; by the time clinical symptoms emerge, kidney damage is typically irreversible. Early detection and precise staging are therefore critical to slow disease progression, reduce complications such as cardiovascular events, and optimise treatment decisions.

Clinical practice guidelines, such as those from the Kidney Disease: Improving Global Outcomes (KDIGO) consortium², recommend staging CKD using both the estimated glomerular filtration rate (eGFR) and albuminuria. The eGFR, calculated from serum creatinine (Scr) via equations such as the CKD Epidemiology Collaboration (CKD-EPI) or the Modification of Diet in Renal Disease (MDRD) formula^{3,4}, remains a cornerstone for diagnosis, medication dosing, and referral timing. The KDIGO classification includes six severity levels (stages 1–5, with stage 3 split into 3a and 3b) that are inherently ordered, and misclassifying between non-adjacent stages can have significant clinical consequences.

Machine learning (ML) offers the potential to improve CKD detection and staging from routine clinical data, yet three persistent challenges limit current approaches:

1. *Stage-aware modelling is underused* Most studies reduce CKD prediction to a binary “disease vs. no disease” task, discarding the ordinal nature of KDIGO staging and increasing the likelihood of large-step misclassifications that matter in practice.

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2. *Physiological plausibility is rarely enforced* Even high-performing models may produce biologically implausible predictions; for example, assigning a milder CKD stage despite higher serum creatinine; undermining clinician trust.
3. *Probability calibration is neglected* Overconfident or poorly calibrated outputs can mislead clinical decision-making, especially when model predictions are integrated into triage or treatment algorithms.

Recent advances in ordinal classification^{5,6} exploit the inherent order of labels to improve discrimination between adjacent categories. Probability calibration techniques, such as temperature scaling^{7,8}, can align predicted probabilities with observed frequencies, reducing the risk of harmful overconfidence. Knowledge distillation^{9,10} enables the transfer of predictive capability from a complex, well-calibrated *teacher* model to a simpler, faster *student* model, suitable for clinical deployment.

Moreover tabular medical data presents a unique challenge: tree-based ensembles (e.g., CatBoost, XGBoost, LightGBM) typically outperform neural networks on small-to-medium datasets due to sample efficiency and native handling of mixed-type features¹¹; however, despite their superior discrimination on tabular benchmarks, these ensemble models lack three properties essential for clinical deployment: i) Differentiability: Tree-based ensembles are piecewise-constant functions that do not support gradient-based constraints. We cannot enforce physiological monotonicity (e.g., $\partial \text{stage} / \partial \text{Scr} \geq 0$) via gradient penalties in CatBoost, whereas neural networks naturally support such constraints through automatic differentiation. ii) Compact deployment: Ensemble models require 100–700 trees, each storing split thresholds, feature indices, and leaf values, resulting in models >100 MB. This is unsuitable for edge devices (mobile health applications, resource-constrained clinics) or real-time inference in clinical workflows where sub-millisecond latency is required. iii) Ordinal structure: Tree-based models treat output classes independently, with no mechanism to encode that CKD stages 1–6 are ordered. This can lead to large-step misclassifications (predicting Stage 1 when truth is Stage 5) that are clinically catastrophic.

Knowledge distillation^{9,12} addresses these limitations by transferring the ensemble's learned patterns into a compact, differentiable neural network. The teacher (calibrated CatBoost) provides rich supervision via soft probability distributions. These softened probabilities expose knowledge about relationships between neighboring CKD stages (e.g., Stage 3a vs 3b are more similar than Stage 1 vs 6); this information would be completely lost if we transferred only hard argmax labels¹³. The student (ordinal neural network) can combine this transferred knowledge with architectural inductive biases: (i) an ordinal head that enforces stage ordering, reducing large-step errors; (ii) an auxiliary eGFR regression task that anchors predictions in continuous kidney function; and (iii) a serum creatinine-monotonicity penalty that ensures physiological consistency. This combination of ensemble discrimination, neural network flexibility, and domain-knowledge constraints forms the foundation of our approach. Used separately, calibration can't fix violations of physical laws; an ordinal structure without calibration yields overconfident predictions. Our integrated framework ensures predictions are simultaneously accurate, well-ordered, calibrated, and physiologically consistent.

To our knowledge, no prior CKD staging framework integrates these three components – (i) ordinal learning, (ii) calibration, and (iii) domain-specific physiological constraints – into a single, coherent approach. In this study, we address these gaps by developing an *informed monotonic neural network* for multi-stage CKD prediction. Our framework combines:

- Ordinal regression (CORN) to respect the KDIGO stage ordering,
- Probability-space knowledge distillation from a calibrated CatBoost teacher to improve generalisation on limited data,
- An auxiliary eGFR regression task to anchor predictions in continuous kidney function,
- A serum creatinine-monotonicity penalty to enforce physiological consistency.

We evaluate our approach on a single-centre registry study of 750 patients from Al-Ramadi Teaching Hospital, Iraq, and compare it against strong baseline models across both discrimination and calibration metrics. Our results demonstrate that integrating clinical priors with modern ML techniques yields stage predictions that are not only accurate but also calibrated and physiologically plausible; these properties are essential for safe and trustworthy AI-assisted CKD management.

Related works

The prediction and staging of chronic kidney disease (CKD) have been widely studied using both traditional machine learning (ML) and deep learning (DL) approaches. The UCI CKD dataset¹⁴, consisting of 400 patient records and 24 clinical attributes, has been the most frequently used benchmark for binary CKD detection. Classical ML models such as logistic regression (LR), decision trees (DT), *k*-nearest neighbours (KNN), support vector machines (SVM), random forests (RF), and gradient boosting (e.g., XGBoost) have consistently achieved high accuracies on this dataset when paired with proper preprocessing and feature selection. Islam et al.¹⁵ compared twelve ML classifiers for early CKD detection, finding XGBoost to achieve 98.33% accuracy, which improved to 99.16% after applying principal component analysis (PCA) for dimensionality reduction.

Sawhney et al.¹⁶ developed a multilayer perceptron (MLP) with feature extraction and selection for CKD detection using the UCI dataset. Their model achieved perfect accuracy on the test set, surpassing LR (96%) and SVM (82%), though the small dataset size raises concerns about overfitting. Alsekait et al.¹⁷ proposed an ensemble deep learning framework combining recurrent neural network (RNN), long short-term memory (LSTM), and gated recurrent unit (GRU) models with an SVM meta-learner. After feature selection via chi-square, recursive feature elimination (RFE), and mutual information, the method achieved 99.69% accuracy, demonstrating the potential of hybrid architectures.

While many works focus on binary detection^{18,19}, several studies have addressed multi-class staging and progression. Rady and Anwar²⁰ used probabilistic neural networks (PNN), MLP, SVM, and radial basis function (RBF) networks on a cleaned subset of the UCI dataset (361 records) for KDIGO-based CKD staging. The PNN achieved 96.7% accuracy, outperforming other models. Xiao et al.²¹ investigated CKD severity prediction in a hospital cohort of 551 patients using nine ML algorithms, including LR, elastic net, RF, and XGBoost. LR achieved the highest area under the ROC curve (AUC) of 0.873; elastic net achieved the highest sensitivity (0.85), and XGBoost obtained the highest specificity (0.83). These results show that well-calibrated linear models can be competitive in clinical settings. Fatima et al.²² developed a novel imputation technique for missing values, and the Deep Neural Network (DNN) improves the CKD model's predictive performance.

A recent systematic review by Iliyas et al.²³ synthesised 86 studies from 2019-2024, concluding that while DL models such as convolutional neural networks (CNNs) and MLPs can capture complex feature interactions, tuned tree-based ensembles like LightGBM, CatBoost, and XGBoost often outperform DL models on small-to-medium tabular datasets due to sample efficiency and interpretability. The review emphasised the importance of patient-level data splitting to avoid leakage, multiclass KDIGO staging (including the 3a/3b subdivision), and explainability techniques such as SHAP and LIME to improve clinical trust.

Ordinal classification methods have been proposed for tasks where class labels are ordered, as in CKD staging. Shi et al.⁵ introduced the CORN framework, which enforces rank consistency through conditional probabilities, achieving superior results to earlier methods such as CORAL⁶ across benchmark datasets. Integrating such approaches in CKD staging helps leverage the natural order of disease stages and reduce misclassifications between adjacent stages.

Model calibration is another critical aspect in medical AI. Guo et al.⁷ demonstrated that modern neural networks tend to be miscalibrated, and that post-hoc temperature scaling can significantly reduce expected calibration error (ECE)²⁴ without affecting accuracy. Proper calibration is particularly important in CKD models to ensure that predicted probabilities reflect true disease likelihoods, supporting safe clinical decision-making.

Knowledge distillation (KD) has been explored to transfer knowledge from complex, high-performing models to simpler or smaller models. Frosst and Hinton²⁵ distilled a deep neural network into a soft decision tree for improved interpretability. Borisov et al.¹⁰ reviewed methods for distilling gradient boosting decision trees (GBDTs) into neural networks for tabular data, showing that probability-based KD can improve student performance on small datasets while retaining efficiency.

Methods
Dataset acquisition and characteristics

A single-centre registry was established at Al-Ramadi Teaching Hospital (Iraq) from January to October 2024 and yielded n = 750 adult encounters. Each record contains fifteen attributes: thirteen candidate predictors and two response/predicted fields (estimated glomerular filtration rate (eGFR), and the physician-assigned CKD stage; full definitions in Table 1). Exclusion criteria included incomplete laboratory profiles, implausible physiological values, and duplicate records.

Missingness was low overall: Glucose 4.0%, Uric acid 2.7%, Serum creatinine 1.3%; all remaining variables were complete. Direct identifiers were stripped at source; the hospital's ethics committee classified the project as non-human-subjects research, waiving informed consent. The de-identified dataset will be made available upon reasonable request under a CC-BY 4.0 licence (see "Data availability"). A histogram of the distribution of CKD stages is shown in Fig. 1.

Attribute	Type	Typical range / levels	Clinical meaning	Missing %
Gender	Nominal	Male, Female	Biological sex	0
Age	Numerical	18-90 yr	Chronological age	0
WBC	Numerical	4-10 × 10 ³ μL ⁻¹	White-blood-cell count	0
RBC	Numerical	4-6.2 × 10 ⁶ μL ⁻¹	Red-blood-cell count	0
HGB	Numerical	11-16 g dL ⁻¹	Haemoglobin concentration	0
HCT	Numerical	37-54 %	Haematocrit	0
PLT	Numerical	150-450 × 10 ³ μL ⁻¹	Platelet count	0
Urea	Numerical	15-45 mg dL ⁻¹	Blood urea nitrogen	0
Serum creatinine (Scr)	Numerical	0.6-1.2 mg dL ⁻¹	Marker of renal clearance	1.3
Uric acid (UA)	Numerical	2.5-6.8 mg dL ⁻¹	Purine-metabolism product	2.7
Glucose (Glu)	Numerical	70-115 mg dL ⁻¹	Random blood glucose	4.0
Hypertension (HRT)	Nominal	Yes, No	Physician-diagnosed hypertension	0
Diabetes mellitus (DM)	Nominal	Yes, No	Physician-diagnosed diabetes	0
eGFR	Numerical	0-120 mL min ⁻¹ (calc.)	MDRD-4 estimate of filtration rate	-
CKD stage	Multiclass	1, 2, 3a, 3b, 4, 5	KDIGO 6-level severity label	-

Table 1. Comprehensive description of the fifteen attributes in the Al-Ramadi CKD cohort. Ranges reflect typical physiological limits. "calc." indicates values derived during pre-processing.

It is important to note that complete KDIGO 2012 staging requires both eGFR categories (G1-G5) and albuminuria categories (A1-A3). Our dataset does not include albuminuria measurements due to limited routine testing at Al-Ramadi Teaching Hospital. Consequently, this study performs eGFR-grade classification along the eGFR axis only, not full KDIGO staging.

Despite being incomplete from a full KDIGO perspective, eGFR-grade classification remains clinically meaningful and practically relevant for four key reasons: i) Real-World Data Availability: Albuminuria testing (urine ACR) is not routinely performed in many healthcare settings. In contrast, serum creatinine is part of standard metabolic panels in >95% of healthcare systems; it is also inexpensive (\$5-10 per test) and widely available. ii) eGFR Dominates Immediate Clinical Decisions: While complete KDIGO staging includes both markers, eGFR is the primary determinant for most immediate clinical actions including medication dosing (antibiotics, anticoagulants, chemotherapy etc.), nephrology referral, dialysis planning, kidney transplant evaluation etc. Albuminuria primarily refines long-term risk stratification for progression and cardiovascular events but does not typically change immediate management for stages G1-G4.

Data preprocessing

Imputation strategy

Rows with more than three missing values did not occur; remaining gaps were imputed once by predictive mean matching (PMM, five donor draws)²⁶. PMM preserves the empirical distribution of skewed laboratory variables more faithfully than mean or median substitution and has been shown to reduce downstream bias compared with deterministic single-value methods²⁷.

Feature scaling and encoding

Continuous predictors were z-standardised using means and standard deviations. Binary predictors (*Gender*, *Hypertension*, *Diabetes mellitus*) were encoded as 0/1.

Exploratory correlation analysis

Spearman correlation coefficients were calculated to identify collinearity (Fig. 2). The strongest positive association was observed between *Urea* and *Serum creatinine* ($\rho = 0.76$), while *Urea* and *Haematocrit* exhibited the strongest negative relationship ($\rho = -0.50$). These findings informed feature-importance interpretation but did not prompt feature removal because tree-based baselines are robust to moderate multicollinearity.

GFR estimation and stage labeling

Accurate estimation of glomerular filtration rate (GFR) is essential because chronic kidney disease (CKD) is often clinically silent until filtration falls well below normal values. Serum creatinine (Scr) (a low-molecular-

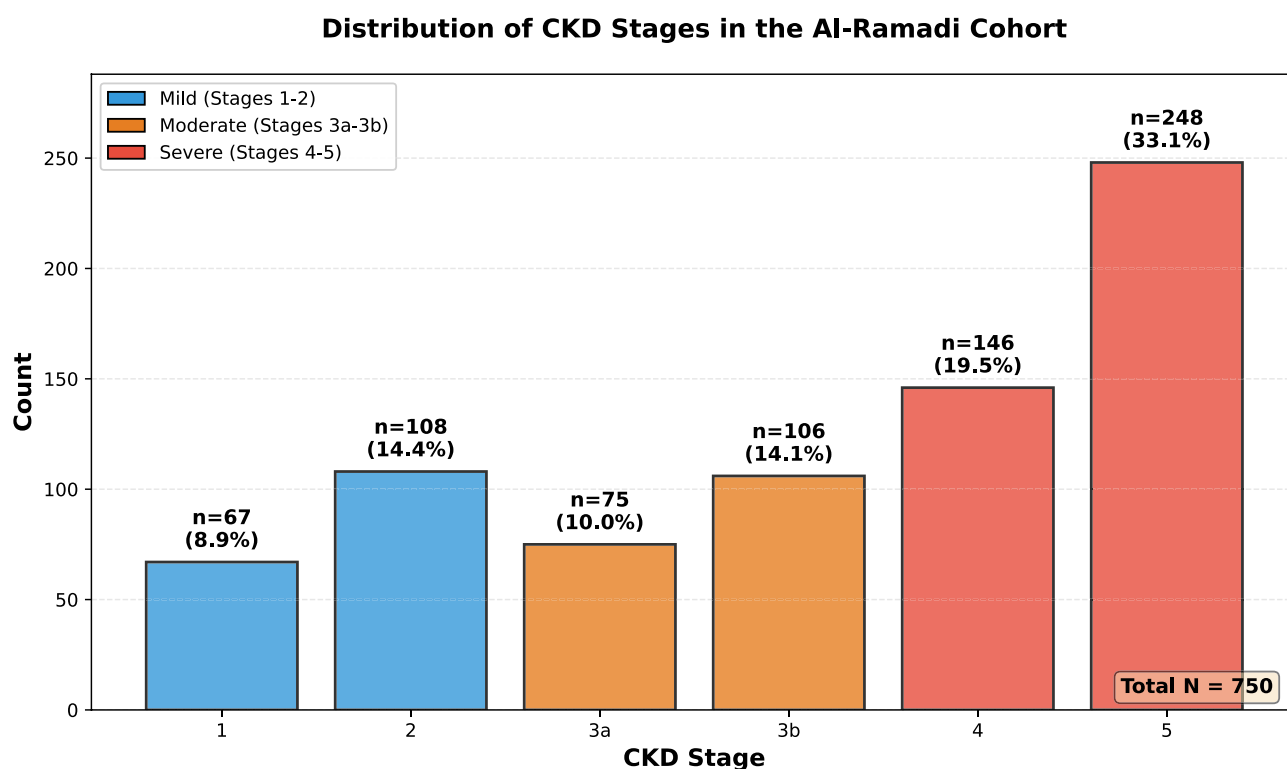


Fig. 1. Distribution of CKD stages in the Al-Ramadi cohort. A histogram of the six KDIGO-defined stages shows the class imbalance in the dataset, providing context for the macro-averaged performance metrics.

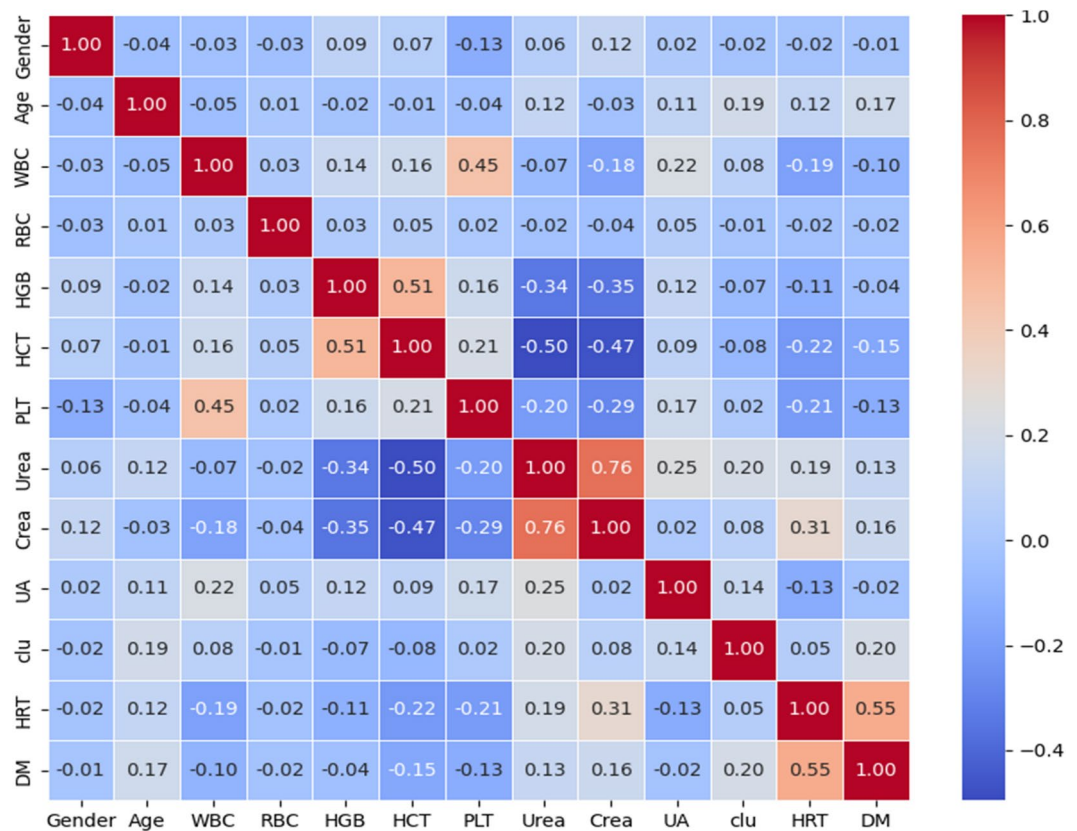


Fig. 2. Feature correlation matrix.

weight waste product cleared almost exclusively by the kidneys) forms the basis of most estimating equations. Two formulae are in common use:

CKD-EPI formula

The revised Chronic Kidney Disease Epidemiology Collaboration equation³ expresses eGFR as

$$\text{eGFR}_{\text{CKD-EPI}} = 142 \times \left[\min\left(\frac{\text{Scr}}{\kappa}, 1\right) \right]^{\alpha} \left[\max\left(\frac{\text{Scr}}{\kappa}, 1\right) \right]^{-1.200} \times 0.9938^{\text{Age}} \times (1.012)^{\text{female}}, \quad (1)$$

with

$$\kappa = \begin{cases} 0.7 & \text{female} \\ 0.9 & \text{male} \end{cases}, \quad \alpha = \begin{cases} -0.241 & \text{female} \\ -0.302 & \text{male} \end{cases},$$

Scr in mg dL^{-1} and Age in years. The min-max construction applies a gentle slope (α) when Scr is below the sex-specific pivot κ and a steeper inverse slope (-1.200) above, thereby reducing proportional error across the physiological Scr range. The coefficient 1.012 accounts for lower baseline creatinine production in women; the 2009 race multiplier has been removed.

MDRD formula

The Modification of Diet in Renal Disease (MDRD) study group proposed an empirically calibrated alternative⁴:

$$\text{eGFR}_{\text{MDRD}} = 175 \times \text{Scr}^{-1.154} \text{Age}^{-0.203} (0.742)^{\text{female}}, \quad (2)$$

where Scr and Age retain the same units. The female coefficient 0.742 corrects for lower muscle-derived Scr; the original African-American term (1.212) is omitted because the present cohort is ethnically homogeneous. Independent evaluations in Middle-Eastern cohorts suggest that MDRD yields smaller bias than CKD-EPI²⁸ at $\text{eGFR} < 60 \text{ mL min}^{-1} 1.73 \text{ m}^2$. Additionally, CKD-EPI 2021 uses slightly different creatinine breakpoints optimized for newer assays; our creatinine values were measured using the Jaffé method standardized to IDMS, which aligns better with MDRD's calibration. We therefore adopted Eq. 2 for all downstream analyses. Computed values were truncated to the physiological range $0\text{--}120 \text{ mL min}^{-1} 1.73 \text{ m}^2$.

Equation 2 was applied row-wise, and resulting values were truncated to $0\text{--}120 \text{ mL min}^{-1} 1.73 \text{ m}^2$ to remove physiologically implausible outliers introduced by imputation. All computations used Scr values traceable

Stage	eGFR range (mL min ⁻¹ 1.73 m ²)	Clinical interpretation	Stage label
1	≥ 90	Normal / hyperfiltration	1
2	60-89	Mild reduction	2
3A	45-59	Mild-moderate reduction	3
3B	30-44	Moderate-severe reduction	4
4	15-29	Severe reduction	5
5	< 15	Kidney failure	6

Table 2. KDIGO 2012 chronic-kidney-disease stage thresholds² applied to derived eGFR values.

Model	Key hyper-parameters (all others = library defaults)
Logistic regression	solver = lbfgs; penalty = ℓ_2 ; $C = 1$; max_iter = 1 000
kNN	$k = 5$; weights = uniform; metric = Minkowski ($p = 2$)
Gaussian naïve Bayes	variance smoothing = 1×10^{-9} (default)
RBF-SVM	kernel = rbf; $C = 1$; γ = scale; probability = True
Random Forest	$n_{\text{trees}} = 100$; criterion = gini; max_depth = None
Extra Trees	$n_{\text{trees}} = 100$; criterion = gini; max_depth = None
Gradient Boosting	$n_{\text{trees}} = 100$; learning_rate = 0.1; max_depth = 3
XGBoost	objective = multi:softprob; $n_{\text{trees}} = 100$; max_depth = 6
LightGBM	objective = multiclass; $n_{\text{trees}} = 100$; num_leaves = 31
CatBoost	iterations = 200; depth = 6; learning_rate = 0.1; loss = MultiClass

Table 3. Hyper-parameters used for the ten reference classifiers. All values are vendor defaults; no grid search or cross-validation was performed.

to IDMS (isotope-dilution mass spectrometry) calibration. Following KDIGO 2012 guidelines², eGFR was discretised into six ordinal CKD stages (Table 2). The transformed stage label replaces the original binary CKD flag and is the primary target for subsequent modelling; treating stage prediction as a six-class classification problem.

Machine learning framework

Baseline learners

To contextualize the design of the proposed informed neural network, we first established a comprehensive suite of baseline classifiers that collectively span the major algorithmic families for tabular data¹¹: linear (logistic regression), distance-based (k NN), probabilistic (Gaussian naïve Bayes), kernel (RBF-SVM) and tree/boosting ensembles (Random Forest, Extra Trees, Gradient Boosting, XGBoost, LightGBM and CatBoost). Hyper-parameters were fixed a-priori as given in Table 3. All baselines are trained once on the full 525-row training set (70% training split with the random seed fixed to 42) and evaluated once on the 225-row hold-out test set; no internal cross-validation is performed. Class weights were not applied at this stage, permitting a fair comparison of raw discrimination capacity. They address the classification task only as continuous eGFR regression is unique to the proposed network and therefore omitted for the comparators. Performance was quantified with a set of scalar measures. For clarity we recall their definitions.

Performance metrics

Let $y_i \in \{1, \dots, 6\}$ denote the true CKD stage label of encounter i and $\hat{y}_i$ the stage predicted by a classifier. The associated class-probability vector is $\mathbf{p}_i = (p_{i1}, \dots, p_{i6})$ with $p_{ik} = \Pr(\text{stage} = k | \mathbf{x}_i)$ and $\sum_{k=1}^6 p_{ik} = 1$. From the predictions of all N test encounters we form the 6×6 confusion matrix $\mathbf{C} = [C_{kj}]$, where C_{kj} counts cases of true class k predicted as j . The following six scalar metrics are derived from $\mathbf{C}$ or $\mathbf{p}_i$.

Accuracy (Acc).

$$\text{Acc} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}\{y_i = \hat{y}_i\}.$$

The overall proportion of correctly classified stages is simple but sensitive to class imbalance.

Balanced accuracy (BalAcc).

$$\text{BalAcc} = \frac{1}{6} \sum_{k=1}^6 \frac{C_{kk}}{\sum_{j=1}^6 C_{kj}}.$$

Computes the per-class recall mean; gives each stage equal influence regardless of prevalence.

Macro-averaged F_1 (Macro- F_1). For each class k

$$\text{Prec}_k = \frac{C_{kk}}{\sum_j C_{jk}}, \quad \text{Rec}_k = \frac{C_{kk}}{\sum_j C_{kj}}, \quad F_{1,k} = \frac{2 \text{Prec}_k \text{Rec}_k}{\text{Prec}_k + \text{Rec}_k}.$$

$$\text{MacroF}_1 = \frac{1}{6} \sum_{k=1}^6 F_{1,k}.$$

Weighs precision and recall equally and averages them over stages; it penalises models that ignore minority classes.

Matthews correlation coefficient (MCC).

$$\text{MCC} = \frac{\sum_{k,j} C_{kk} C_{jj} - \sum_{k,j} C_{kj} C_{jk}}{\sqrt{\left(\sum_k r_k\right) \left(\sum_j c_j\right) - \sum_k r_k^2} \sqrt{\left(\sum_k c_k\right) \left(\sum_j r_j\right) - \sum_k c_k^2}},$$

where $r_k = \sum_j C_{kj}$ and $c_k = \sum_j C_{jk}$. Yields a single chance-corrected correlation that remains interpretable under class imbalance.

Macro area under the ROC curve (Macro-AUC). For each class k a one-versus-rest ROC curve is traced using $\{p_{ik}\}_{i=1}^N$; the six areas are averaged:

$$\text{MacroAUC} = \frac{1}{6} \sum_{k=1}^6 \text{AUC}_k.$$

Evaluates ranking ability independent of any decision threshold.

Brier score.

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N \sum_{k=1}^6 (p_{ik} - \mathbb{1}_{\{y_i = k\}})^2.$$

Mean-squared deviation between predicted probabilities and one-hot targets; lower values indicate better probabilistic calibration.

Accuracy and Balanced-accuracy quantify hard-label discrimination, with and without prevalence correction. Macro- F_1 rewards balanced precision-recall across stages, essential for equitable clinical triage. MCC provides a robust, chance-adjusted summary unaffected by class imbalance. Macro-AUC measures threshold-free ranking performance, relevant when operating points differ between screening and specialist contexts. Finally, the Brier score assesses probability calibration-critical because well-calibrated probabilities underpin the temperature-softening step used later in knowledge distillation. Together these six metrics furnish a comprehensive and clinically meaningful yard-stick for comparing baseline models with the proposed informed monotone network.

Statistical uncertainty quantification

All performance metrics are reported with 95% confidence intervals computed using stratified bootstrap resampling. We draw $B = 1000$ bootstrap samples with replacement from the 225-sample test set, preserving class proportions in each sample, and compute all metrics on each bootstrap replicate. Confidence intervals are constructed using the percentile method: the 2.5th and 97.5th percentiles of the bootstrap distribution form the lower and upper bounds, respectively²⁹.

This approach accounts for sampling variability in the fixed test set and provides valid uncertainty estimates under mild regularity conditions. We chose 1000 iterations to ensure stable percentile estimates, following standard recommendations²⁹.

As seen in Table 4, among ensembles, CatBoost achieved the best discrimination and calibration (Macro-AUC 0.995; Brier 0.045); hence, we use it as the teacher in the distillation pipeline.

Teacher-student knowledge distillation

The knowledge-distillation (KD) pipeline relies on a *teacher* that produces well-calibrated, information-rich class probabilities. We first motivate the choice of distillation strategy, then specify the teacher configuration and calibration, and finally detail what is passed to the student during training.

Teacher-student knowledge distillation

KD methods are commonly grouped by their *knowledge source* (logits (transfers the teacher's class probabilities), feature maps (forces the student to mimic intermediate feature maps), relational graphs (preserves pairwise distances or angles between samples in the teacher's embedding space)), the *transfer scheme* (teacher $\rightarrow$ student, self-distillation, mutual), and the *training paradigm* (offline, on-the-fly, data-free)^{30,31}. Feature/relational KD can outperform the original logit-based method¹² on large image datasets or NLP benchmarks, but they require either *structurally compatible hidden activations* or *millions of synthetic samples*; neither of which is available in

Model	Acc	Bal-Acc	Macro-F1	MCC	Macro-AUC	Brier
Logistic regression	0.780	0.696	0.702	0.725	0.939	0.169
	[0.725, 0.829]	[0.638, 0.751]	[0.645, 0.756]	[0.672, 0.775]	[0.918, 0.956]	[0.142, 0.198]
kNN (5)	0.560	0.505	0.501	0.456	0.835	0.218
	[0.496, 0.622]	[0.445, 0.565]	[0.442, 0.560]	[0.395, 0.516]	[0.802, 0.865]	[0.185, 0.253]
Gaussian naïve Bayes	0.620	0.567	0.489	0.544	0.930	0.287
	[0.556, 0.680]	[0.505, 0.627]	[0.425, 0.551]	[0.485, 0.602]	[0.908, 0.948]	[0.251, 0.325]
RBF-SVM	0.653	0.561	0.568	0.564	0.920	0.167
	[0.591, 0.713]	[0.500, 0.621]	[0.508, 0.627]	[0.505, 0.622]	[0.895, 0.941]	[0.141, 0.195]
Random Forest	0.853	0.807	0.811	0.816	0.975	0.146
	[0.807, 0.893]	[0.759, 0.850]	[0.765, 0.853]	[0.772, 0.857]	[0.962, 0.985]	[0.118, 0.177]
Extra Trees	0.840	0.808	0.816	0.799	0.963	0.173
	[0.793, 0.882]	[0.761, 0.852]	[0.771, 0.858]	[0.754, 0.841]	[0.947, 0.976]	[0.143, 0.205]
Gradient Boosting	0.900	0.875	0.869	0.875	0.978	0.080
	[0.858, 0.938]	[0.833, 0.913]	[0.828, 0.907]	[0.835, 0.912]	[0.966, 0.987]	[0.061, 0.102]
XGBoost	0.900	0.871	0.868	0.875	0.987	0.068
	[0.858, 0.938]	[0.828, 0.910]	[0.827, 0.906]	[0.835, 0.912]	[0.977, 0.993]	[0.051, 0.088]
LightGBM	0.920	0.897	0.897	0.900	0.988	0.065
	[0.882, 0.953]	[0.858, 0.932]	[0.858, 0.932]	[0.863, 0.934]	[0.979, 0.994]	[0.048, 0.085]
CatBoost	0.940	0.933	0.926	0.925	0.995	0.045
	[0.907, 0.969]	[0.898, 0.963]	[0.891, 0.957]	[0.891, 0.956]	[0.990, 0.998]	[0.032, 0.061]

Table 4. Overall classification performance of baseline learners on the 225-row hold-out test set. Bold entries mark the best value in each column. Acc = accuracy; Bal-Acc = balanced accuracy; MCC = Matthews correlation coefficient. 95% confidence intervals computed using stratified bootstrap (1000 iterations). Intervals shown in brackets [lower, upper] below point estimates.

our case because our dataset contains only 525 training encounters; allocating GPU memory to feature tensors or pairwise graphs would crowd out the data itself. Logit-based distillation with temperature scaling, by contrast, needs no additional memory and has outperformed heavier schemes for small, tabular, mixed-type data^{32,33}. We therefore adopt logit KD.

Teacher model selection and calibration

Among the baselines (Table 4), CatBoost achieved the highest Macro-AUC (0.995) and lowest Brier score (0.045), indicating superior calibration. CatBoost grows *oblivious* (symmetric) trees and handles categorical variables via ordered target statistics³⁴.

Raw tree probabilities can be over-confident. We calibrate with a single temperature T on a fixed 15 % split by minimising multiclass negative log-likelihood (NLL):

$$T^* = \arg \min_{T>0} - \sum_{i=1}^{N_{\text{cal}}} \log \left(\frac{p_{i y_i}^{1/T}}{\sum_k p_{i k}^{1/T}} \right).$$

The optimal $T^* = 3$ reduced NLL from 0.612 to 0.426. We also found that the expected calibration error (ECE) using 10 equal-width bins, represented as:

$$\text{ECE} = \sum_{b=1}^{10} \frac{n_b}{N} |\text{acc}(b) - \text{conf}(b)|,$$

where n_b is the number of samples in bin b , $\text{acc}(b)$ is the empirical accuracy and $\text{conf}(b)$ the mean predicted confidence in that bin. After temperature scaling the ECE dropped from 4.7 % to 3.1 %, confirming that the teacher's probabilities are well calibrated for knowledge transfer.

ECE quantifies the difference between predicted confidence and empirical accuracy across probability bins³⁵. Traditional ECE uses fixed-width binning (10 equal intervals in $[0,1]$), but this can produce unstable estimates with small sample sizes when some bins contain few or no samples³⁶. We compute ECE using three binning strategies: i) Fixed-width ECE (as defined above) ii) Equal-frequency ECE: Bins are defined by quantiles of the predicted confidence distribution such that each bin contains approximately n/M samples. This approach is more stable with modest sample sizes as it avoids empty or sparsely populated bins³⁶. We use $M = 10$ bins for aggregate ECE and $M = 5$ bins for class-wise ECE (due to smaller per-class samples). iii) Equal-mass ECE: Similar to equal-frequency but bins are weighted by the empirical confidence distribution, providing additional robustness to outliers.

Class-wise calibration

Aggregate ECE can mask class-specific miscalibration. A model may appear well-calibrated overall while being severely miscalibrated for individual classes, which is problematic for medical decision-making where different stages have different clinical implications.

For each CKD stage $k \in \{1, \dots, 6\}$, we compute one-vs-rest binary calibration: i) Confidence: $c_i = P(y_i = k | \mathbf{x}_i)$ (predicted probability for stage k) ii) Correctness: $\mathbb{1}[y_i = k]$ (binary indicator) iii) Binning: Equal-frequency with $M = 5$ bins per class iv) Bootstrap: 1000 iterations for 95% confidence intervals

Brier score decomposition

The Brier score can be decomposed into three orthogonal components using the Murphy decomposition³⁷:

$$\text{BS} = \text{Reliability} - \text{Resolution} + \text{Uncertainty} \quad (3)$$

where; Reliability is the average squared deviation between predicted probabilities and observed frequencies (miscalibration penalty). Lower is better; 0 indicates perfect calibration. Resolution is the ability to discriminate between outcomes by assigning different probability values (sharpness of predictions). Higher is better; captures predictive power. Uncertainty is the inherent unpredictability due to class prior distribution. This is constant for a given dataset and cannot be improved by the model.

This decomposition reveals whether Brier score differences reflect calibration issues (reliability term) or discriminative power (resolution term), providing deeper insight than the aggregate score alone. We report all components with 95% bootstrap confidence intervals (1000 iterations).

Soft-probability distillation (what the student sees)

For encounter i the teacher outputs $\mathbf{p}_i = (p_{i1}, \dots, p_{i6})$ whose largest component corresponds to the teacher's predicted CKD stage. Before these probabilities are used as a training signal they are *softened* with the calibrated temperature $T^* = 3$. We form temperature-softened targets as:

$$\tilde{p}_{ik} = \frac{p_{ik}^{1/T^*}}{\sum_{j=1}^6 p_{ij}^{1/T^*}}, \quad T^* = 3.$$

The exponent $1/T^* < 1$ flattens the distribution; increasing the weight of classes, exposing the knowledge about the relationships between neighbouring CKD stages that would be lost if only the arg-maxlabel were transferred¹³. Because the student's CORN head outputs probabilities directly (explored in the next section), distillation is performed *in probability space*. Let i index encounters ($i = 1, \dots, N$) and let $\mathcal{K} = \{1, \dots, 6\}$ be the set of stages. For encounter i ,

- the teacher's *raw* probability vector is $\mathbf{p}_i = \langle p_{ik} \rangle_{k \in \mathcal{K}}, \sum_k p_{ik} = 1$;
- the *softened* teacher vector is $\tilde{\mathbf{p}}_i = \langle \tilde{p}_{ik} \rangle_{k \in \mathcal{K}}$ with

$$\tilde{p}_{ik} = \frac{p_{ik}^{1/T^*}}{\sum_{j \in \mathcal{K}} p_{ij}^{1/T^*}}, \quad T^* = 3;$$

- the student's CORN head outputs a probability vector $\hat{\mathbf{p}}_i = \langle \hat{p}_{ik} \rangle_{k \in \mathcal{K}}, \sum_k \hat{p}_{ik} = 1$.

Because the student already produces probabilities, distillation is performed *in probability space* using the Kullback-Leibler divergence, averaged over the batch:

$$\mathcal{L}_{\text{KD}} = \frac{1}{N} \sum_{i=1}^N \text{KL}(\tilde{\mathbf{p}}_i \| \hat{\mathbf{p}}_i) = \frac{1}{N} \sum_{i=1}^N \sum_{k \in \mathcal{K}} \tilde{p}_{ik} \log \frac{\tilde{p}_{ik}}{\max(\hat{p}_{ik}, 10^{-8})}.$$

No additional scaling of student logits and no T^2 gradient correction are required (those are only needed for logit-space KD).

Why heavier KD variants were not used. Feature matching (FitNets³⁸, Attention Transfer³⁹) presupposes convolutional/transformer backbones and inflates the student, while relational/contrastive KD scales as $O(N^2)$ in memory untenable alongside per-batch MixUp on this dataset^{30,40}. We also trialled ordinal focal-KL and a 5-fold CatBoost ensemble; both raised validation Macro-AUC by only 0.002 at (4-5)× compute and yielded no significant test-set gain. We therefore retain single-temperature logit KD.

Student network architecture

The student is a compact, fully differentiable *residual MLP* that ingests the same 13 predictors as the baselines and emits (i) a probability distribution over six CKD stages and (ii) a scalar eGFR (GRF estimate). The topology satisfies the constraint of exact differentiability with respect to Scr (serum creatinine) to enable a gradient-based *creatinine-monotone* penalty in the final loss function.

Let $\mathcal{K} = \{1, \dots, 6\}$ be the set of CKD stages. A mini-batch contains B encounters $(\mathbf{x}_i)_{i=1}^B$, each $\mathbf{x}_i \in \mathbb{R}^{13}$. The teacher is a fixed CatBoost model $\mathcal{T}$; $T^*=3$ is the calibrated temperature. The student has width H , depth

D , dropout p (disabled at evaluation), and parameters θ . Algorithm 1 depicts the forward pass of the teacher softening and residual-MLP student inference.

Input: $(\mathbf{x}_i)_{i=1}^B$; teacher $\mathcal{T}$; temperature T^* ; student params θ
Output: Teacher soft targets $(\tilde{\mathbf{p}}_i)_{i=1}^B$; student probs $(\hat{\mathbf{p}}_i)_{i=1}^B$; eGFR $(\hat{g}_i)_{i=1}^B$; predicted labels $(\hat{y}_i)_{i=1}^B$

Teacher probabilities and temperature softening:

```

for  $i = 1$  to  $B$  do
   $\mathbf{p}_i \leftarrow \mathcal{T}(\mathbf{x}_i)$  // CatBoost raw probs;  $\sum_k p_{ik} = 1$ 
  for  $k \in \mathcal{K}$  do
     $\tilde{p}_{ik} \leftarrow \frac{(p_{ik})^{1/T^*}}{\sum_{j \in \mathcal{K}} (p_{ij})^{1/T^*}}$ 
  end
end

```

Student residual-MLP:

```

for  $i = 1$  to  $B$  do
   $h \leftarrow \text{ReLU}(W_0 \mathbf{x}_i + b_0)$  // stem,  $W_0 \in \mathbb{R}^{H \times 13}$ 
  for  $\ell = 1$  to  $D$  do
     $u \leftarrow \text{ReLU}(W_{\ell,1} h + b_{\ell,1})$ 
     $u \leftarrow \text{Dropout}_p(u)$  // disabled at eval
     $h \leftarrow \text{ReLU}(h + W_{\ell,2} u + b_{\ell,2})$  // residual add
  end
   $h_D \leftarrow h$ 
  Student CORN ordinal head (probabilities):
   $a \leftarrow \sigma(W_{\text{ord}} h_D + b_{\text{ord}}) \in (0, 1)^5$  // logistic
   $c_1 \leftarrow a_1$ ;  $c_k \leftarrow c_{k-1} \cdot a_k$  for  $k = 2..5$  // cumulative product

   $\hat{\mathbf{p}}_i = (p_{i1}, \dots, p_{i6})$  with  $p_{i1} = 1 - c_1$ ,  $p_{ik} = c_{k-1} - c_k$  ( $k = 2..5$ ),  $p_{i6} = c_5$ 
  //  $\sum_k \hat{p}_{ik} = 1$ ; clamp to  $\geq 10^{-8}$ 

  Student eGFR head (linear):
   $\hat{g}_i \leftarrow w_{\text{reg}}^\top h_D + b_{\text{reg}}$ 

  Predicted label (for confusion matrix later):
   $\hat{y}_i \leftarrow \arg \max_{k \in \mathcal{K}} \hat{p}_{ik}$ 
end

```

return $\{(\tilde{\mathbf{p}}_i), (\hat{\mathbf{p}}_i), (\hat{g}_i), (\hat{y}_i)\}_{i=1}^B$

Algorithm 1. Forward pass: teacher softening and residual-MLP student inference

The network consists of a *stem*, a depth- D *residual trunk*, and two *task heads*. All non-linearities are ReLU; dropout with rate p is applied only inside the residual branch. Figure 3 illustrates the complete architecture.

Network topology

Stem (input projection). The stem is a single affine transformation followed by a ReLU nonlinearity: $h_0 = \text{ReLU}(W_0 \mathbf{x}_i + b_0)$ with $W_0 \in \mathbb{R}^{H \times 13}$. Its role is to map the 13 standardised predictors into a higher-dimensional feature space of width H where simple linear separations become more feasible. A single wide projection keeps the parameter count low while providing enough capacity for downstream residual blocks to refine the representation. ReLU is chosen for its piecewise-linear behaviour, which preserves exact input-gradient computations needed by the creatinine-monotone penalty.

Residual trunk (feature refinement with stable gradients). The trunk comprises D residual blocks; each block applies a small two-layer transformation to $h_{\ell-1}$ and adds it back to form h_ℓ :

$$u_\ell = \text{ReLU}(W_{\ell,1} h_{\ell-1} + b_{\ell,1}), \quad u_\ell = \text{Dropout}_p(u_\ell),$$

$$h_\ell = \text{ReLU}(h_{\ell-1} + W_{\ell,2} u_\ell + b_{\ell,2}).$$

The residual connection (identity skip) helps optimisation by allowing gradients to flow unimpeded through depth, reducing the risk of vanishing updates on this small tabular dataset. Dropout is placed *inside* the residual branch only; this regularises the learnable transformation while leaving the skip path deterministic, which

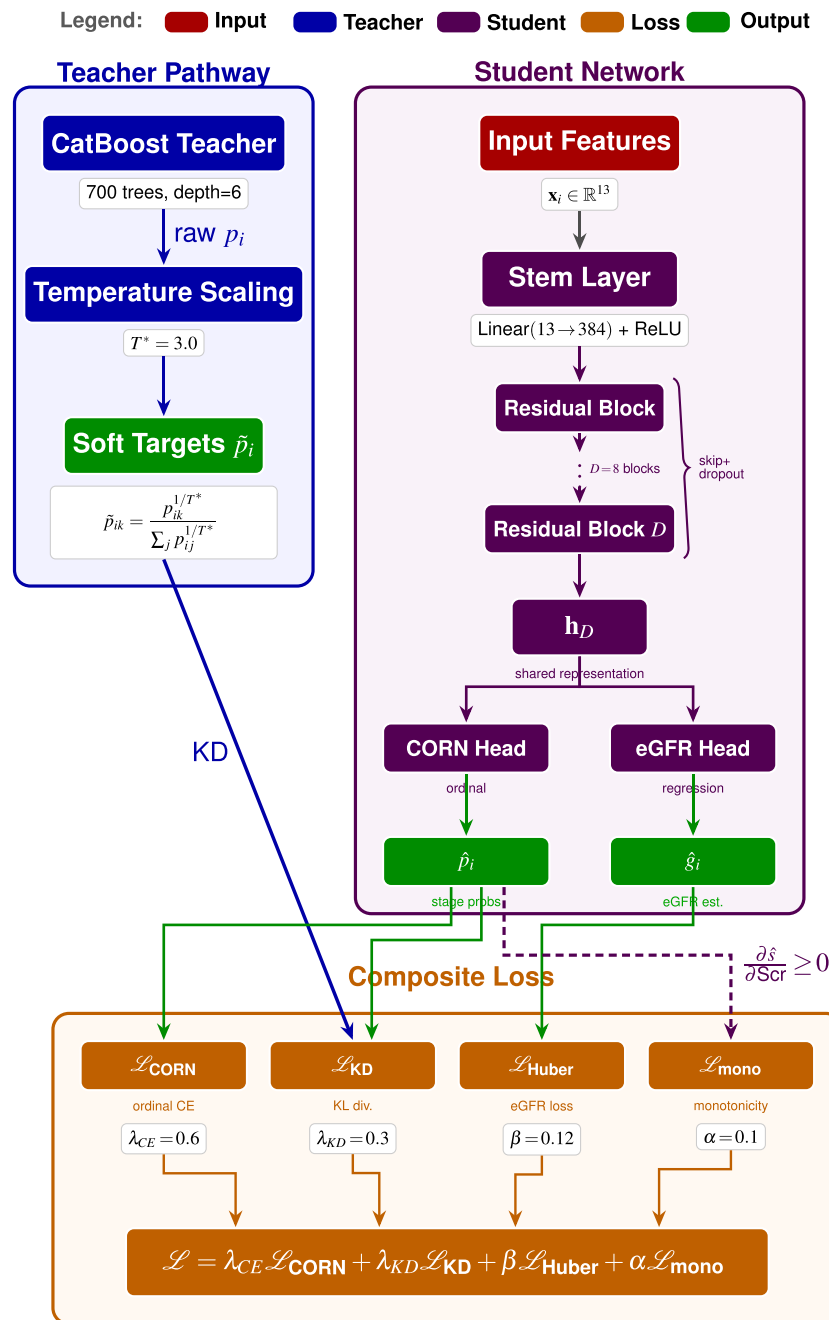


Fig. 3. Architecture of the proposed informed monotone neural network with knowledge distillation. Teacher Pathway: A calibrated CatBoost ensemble (700 trees, depth 6) produces probability distributions softened via temperature scaling ($T^* = 3.0$) to generate soft targets $\tilde{p}_i$. Student Network: Input features $\mathbf{x}_i \in \mathbb{R}^{13}$ pass through a stem layer and $D = 8$ residual blocks (with skip connections and dropout), producing shared representation h_D . Two task heads branch from h_D : a CORN ordinal classifier outputting stage probabilities $\hat{p}_i$, and a linear head for eGFR estimation $\hat{g}_i$. Composite Loss: $\mathcal{L}_{\text{CORN}}$ (ordinal CE, $\lambda_{\text{CE}} = 0.6$), $\mathcal{L}_{\text{KD}}$ (KL divergence, $\lambda_{\text{KD}} = 0.3$), $\mathcal{L}_{\text{Huber}}$ (eGFR loss, $\beta = 0.12$), and $\mathcal{L}_{\text{mono}}$ (monotonicity penalty enforcing $\partial \hat{s} / \partial \text{Scr} \geq 0$, $\alpha = 0.1$).

stabilises both training and the input-output derivatives used by the monotone penalty. We deliberately omit batch normalisation: small mini-batches and mixed-type tabular features often make batch statistics noisy, which can degrade calibration and shift train-test behaviour.

Ordinal classification head (CORN mapping). CKD stages are inherently ordered. Instead of emitting six independent logits with a softmax, the classifier uses a CORN-style ordinal head with $K-1 = 5$ learned thresholds. A linear layer followed by logistic activations produces $a = \sigma(W_{\text{ord}} h_D + b_{\text{ord}}) \in (0, 1)^5$. Cumulative probabilities $c_k = \prod_{j=1}^k a_j$ define the class probabilities

$$\hat{p}_{i1} = 1 - c_1, \quad \hat{p}_{ik} = c_{k-1} - c_k \quad (k = 2, \dots, 5), \quad \hat{p}_{i6} = c_5, \quad \sum_{k=1}^6 \hat{p}_{ik} = 1.$$

This construction embeds the ordering directly into the output layer and has two practical effects: (i) it improves data efficiency when middle stages are under-represented; and (ii) it yields calibrated probabilities that align well with probability-space distillation from the teacher. For numerical stability we clamp probabilities below 10^{-8} inside the loss.

It is important to note that standard multiclass classification with softmax cross-entropy treats all K classes as categorically independent which is, predicting Stage 1 when the true label is Stage 6 incurs the same loss as predicting Stage 5. This independence assumption ignores the ordinal structure of CKD staging, where adjacent stages share clinical similarity and distant misclassifications carry greater clinical consequence. In contrast, CORN models the cumulative probability structure by learning $K - 1$ binary classifiers for the conditional probabilities $P(Y > k | Y > k - 1, \mathbf{x})$, which are then composed via the chain rule to obtain class probabilities. Specifically, the network outputs $a_k = \sigma(\mathbf{w}_k^\top \mathbf{h}_D + b_k)$ representing the probability of exceeding threshold k given that threshold $k - 1$ was exceeded; the cumulative products $c_k = \prod_{j=1}^k a_j$ encode rank-consistency by construction, since predicting a higher stage requires “passing through” all intermediate thresholds. The ordinal negative log-likelihood (defined in the Physics-informed objective section) then penalizes predictions that violate this cumulative structure, concentrating gradients on the decision boundaries between adjacent stages rather than treating all misclassifications uniformly. This formulation provides two key advantages: i) it reduces large-step errors because the model cannot predict Stage 5 without first predicting high probabilities for exceeding Stages 1–4, and ii) it improves sample efficiency for minority middle stages by leveraging shared information across neighboring thresholds.

eGFR regression head. A linear readout $\hat{g}_i = \mathbf{w}_{\text{reg}}^\top \mathbf{h}_D + b_{\text{reg}}$ shares the trunk representation to predict eGFR. The head is deliberately linear: it adds a physiology-informed target without introducing extra non-linearities that could make input-gradient behaviour brittle. During training, only encounters with finite MDRD-derived targets contribute to the Huber loss; otherwise the term is skipped. This auxiliary task encourages the shared features to track renal function continuously, which supports more consistent stage boundaries.

Differentiability and creatinine-monotone penalty. Define the expected stage $\hat{s}(\mathbf{x}_i) = \sum_{k=1}^6 k \hat{p}_{ik}$. Because every operation from $\mathbf{x}_i$ to $\hat{p}_i$ is piecewise linear or smooth, $\partial \hat{s} / \partial \text{Scr}$ exists almost everywhere and is computed exactly by automatic differentiation. The physics-informed loss penalises negative values of this derivative, enforcing that (*ceteris paribus*) increasing serum creatinine should not reduce the predicted stage. Placing dropout only in the residual branch and avoiding batch normalisation ensures these gradients remain stable and interpretable.

Compatibility with teacher distillation. The CORN head returns probabilities directly, $\hat{p}_i$. We therefore match the teacher’s temperature-softened probabilities $\tilde{p}_i$ using a KL divergence in probability space, $\text{KL}(\tilde{p}_i \| \hat{p}_i)$. No logit division by T^* or T^2 rescaling is required (those are specific to logit-space KD). This keeps the optimisation landscape well conditioned and aligns the ordinal output with the calibrated teacher targets.

Parameterisation, complexity, and initialisation. The total parameters are

$$\underbrace{H(13+1)}_{\text{stem}} + \underbrace{D \cdot 2(H^2 + H)}_{\text{two } H \times H \text{ linears per block}} + \underbrace{(K-1)(H+1)}_{\text{ordinal head}} + \underbrace{(H+1)}_{\text{regression head}}.$$

corresponding to the stem, two linear layers per residual block, the ordinal head, and the regression head. With $(H, D, K) = (384, 8, 6)$ this is $\approx 2.37 \times 10^6$ parameters (~ 9.5 MB in fp32). Per sample, the dominant multiply-accumulate cost is $\mathcal{O}(DH^2)$ (here ≈ 2.36 M MACs), supporting sub-millisecond inference on a modern GPU (we used NVIDIA GeForce RTX 5090) and interactive latency on CPU. Weights use He (Kaiming) normal initialisation and biases are zero, which is appropriate for ReLU networks and helps maintain signal variance through depth.

Regularisation and hyper-parameters. Dropout rate p , width H , and depth D are selected once with Optuna⁴¹ on the training partition; chosen values are reported in the Training subsection. If required by class imbalance, per-class weights are applied within the ordinal cross-entropy; these are fixed a priori to avoid target leakage. At evaluation time, dropout is disabled, yielding deterministic predictions and stable calibration curves.

Physics-informed objective

The student is trained with a multi-term objective that balances four signals: (i) *ordinal discrimination* of CKD stages, (ii) *knowledge transfer* from a calibrated CatBoost teacher, (iii) a *physiological auxiliary task* (eGFR regression), and (iv) a *monotonicity prior* enforcing that higher serum creatinine (Scr) should not reduce the predicted stage. This combination aims to retain the teacher’s tabular prowess while steering the network toward clinically consistent behaviour.

For a mini-batch $\mathcal{B}$ of size B , let $\mathbf{x}_i \in \mathbb{R}^{13}$ be the standardised features and $y_i \in \{1, \dots, 6\}$ the true stage. The student outputs a probability vector $\hat{p}_i = (\hat{p}_{i1}, \dots, \hat{p}_{i6})$ with $\sum_k \hat{p}_{ik} = 1$, and an eGFR estimate $\hat{g}_i \in \mathbb{R}$. The teacher provides temperature-softened probabilities $\tilde{p}_i$ (see Teacher subsection). We denote the index of Scr in $\mathbf{x}_i$ by i_{Scr} .

(1) Ordinal discrimination (CORN negative log-likelihood). Because CKD stages are ordered, the classifier uses a CORN head that yields probabilities directly (Methods-Student architecture). The primary classification term is the weighted negative log-likelihood

$$\mathcal{L}_{\text{CORN}} = -\frac{1}{B} \sum_{i \in \mathcal{B}} w_{y_i} \log(\max(\hat{p}_{i y_i}, 10^{-8})),$$

with per-class weights w_k fixed a priori to mitigate imbalance (default $w = [1, 1, 1.8, 1, 1, 1]$).

(2) Knowledge transfer (probability-space distillation). To expose knowledge about non-argmax classes, we match the student probabilities to the teacher's softened targets via a KL divergence computed *in probability space*:

$$\mathcal{L}_{\text{KD}} = \frac{1}{B} \sum_{i \in \mathcal{B}} \text{KL}(\tilde{p}_i \| \hat{p}_i) = \frac{1}{B} \sum_{i \in \mathcal{B}} \sum_{k=1}^6 \tilde{p}_{ik} \log \frac{\tilde{p}_{ik}}{\max(\hat{p}_{ik}, 10^{-8})}.$$

Because the student already outputs probabilities (CORN), no division of logits by T^* and no T^2 gradient scaling are required.

(3) Physiology auxiliary task (eGFR regression). The regression head predicts $\hat{g}_i$ and is trained with a Huber loss on encounters with finite MDRD-derived targets:

$$\mathcal{L}_{\text{Huber}} = \frac{1}{|M|} \sum_{i \in M} \text{Huber}(\hat{g}_i, g_i^*), \quad M = \{i \in \mathcal{B} : g_i^* \text{ is finite}\}.$$

If $|M| = 0$ in a batch, this term is set to zero. The eGFR task encourages the shared representation to track renal function continuously.

(4) creatinine-monotonicity (gradient penalty). Let the expected stage be

$$\hat{s}(x_i) = \sum_{k=1}^6 k \hat{p}_{ik}.$$

We obtain the input gradient $\partial \hat{s} / \partial x_i$ by automatic differentiation and penalise decreases of $\hat{s}$ when Scr increases:

$$\mathcal{L}_{\text{mono}} = \frac{1}{B} \sum_{i \in \mathcal{B}} \max\left(0, -\frac{\partial \hat{s}}{\partial x_i} \Big|_{i_{\text{Scr}}}\right).$$

This local (differential) constraint implements the clinical prior that, *ceteris paribus*, higher creatinine should not imply a milder stage. Using standardised inputs makes the penalty naturally scale-free.

It is important to be precise about what $\mathcal{L}_{\text{mono}}$ enforces. The penalty operates on the local (gradient-based) derivative as for each training sample x_i , it penalises configurations of the network weights under which $\frac{\partial \hat{s}}{\partial x_i} < 0$, i.e., configurations in which an infinitesimally small increase in Scr at x_i would decrease the predicted expected stage. The penalty is therefore a pointwise, first-order condition enforced at the 525 training points encountered across mini-batches. This is distinct from global monotonicity, which would require $\hat{s}(x + \delta e_{\text{Scr}}) \geq \hat{s}(x)$ (where e_{Scr} is the unit vector along the Scr feature dimension) for all x in the input domain and all $\delta > 0$. This is a function-class property; as it holds for every possible combination of the other 12 features and for every positive perturbation magnitude, not just the training points or finite test perturbations. Architectures that achieve global monotonicity by construction include: i) Lattice networks⁴²: here, the function is represented as a piecewise-linear interpolation over a multi-dimensional grid; the interpolation is guaranteed to be monotone if all lattice-vertex parameters are constrained to be non-decreasing in the relevant dimension. ii) Min-max networks⁴³: here, outputs are composed exclusively of monotone operations (max, min, addition, ReLU with non-negative weights); the composition of monotone operations is monotone by the chain rule of monotonicity. iii) Monotone neural networks with constrained weights⁴⁴: these are networks in which all weights along a designated input dimension are forced non-negative and all activations are non-decreasing; by induction, the network output is non-decreasing in that input.

Our residual MLP does none of these as weights in the residual blocks can be negative, and the skip connection can amplify or suppress changes to Scr in non-monotone ways. The gradient penalty steers the learned function toward a region of weight space where local monotonicity holds at training points, but it does not restructure the function class itself.

Local enforcement does not preclude monotonicity violations at unsampled points in the input space, at perturbation magnitudes beyond the infinitesimal approximation, or along joint trajectories that simultaneously change Scr and other features. Global monotonicity can be guaranteed architecturally; for example, by lattice networks or min-max networks (see Limitations); but these approaches incur the curse of dimensionality when applied across all 13 features and restrict the function class in ways that may harm discrimination against minority stages. We therefore adopt local enforcement as the primary mechanism and complement it with the finite-difference audit described in the Results (Table 20), which empirically quantifies the extent to which local enforcement translates to practical, finite-step monotonicity at held-out test points.

Composite objective and schedules. The total loss for one mini-batch is

$$\mathcal{L} = \lambda_{\text{CE}} \mathcal{L}_{\text{CORN}} + \lambda_{\text{KD}} \mathcal{L}_{\text{KD}} + \beta \mathcal{L}_{\text{Huber}} + \alpha_{\text{crea}}(t) \mathcal{L}_{\text{mono}} \quad (4)$$

with $\lambda_{\text{CE}} = 0.6$ fixed at the start and the remaining weights tuned on the training split. To avoid over-constraining early learning, the creatinine penalty follows a ramp:

$$\alpha_{\text{crea}}(t) = \alpha_{\text{final}} \cdot \rho(t), \quad \rho(t) = \gamma \cdot \min\left(1, \frac{t}{400}\right),$$

where t is the epoch index and γ set to 2.4 at the start. This schedule allows the classifier to discover a coarse decision structure before the monotone prior is enforced strongly.

The creatinine-monotonicity constraint is a direct encoding of the physiological law that defines the outcome variable. Serum creatinine is cleared almost exclusively by glomerular filtration (tubular reabsorption and secretion are negligible in the clinical range) so plasma creatinine is inversely proportional to GFR i.e. $\text{Scr} \propto 1/\text{GFR}$. Both eGFR estimation equation used in this study (MDRD) encode this relationship explicitly through Scr raised to a negative exponent. Since the KDIGO staging system assigns stages G1-G5 based solely on eGFR thresholds (G1: ≥ 90 ; G2: 60-89; G3a: 45-59; G3b: 30-44; G4: 15-29; G5: <15 mL/min/1.73 m^2), the monotone direction higher Scr $\Rightarrow$ lower eGFR $\Rightarrow$ higher stage is not an empirical observation derived from the training data but a mathematical consequence of the definitions. A model that predicts a lower CKD stage for a patient with Scr = 5.0 mg/dL than for one with Scr = 2.0 mg/dL (all other features equal) is definitionally inconsistent with the KDIGO system. Enforcing $\delta \hat{s} / \delta \text{Scr} \geq 0$ therefore imposes not a prior belief but a hard biological fact. Moreover, since the eGFR auxiliary regression target is itself a deterministic function of Scr, the monotonicity penalty is reinforced by and consistent with the Huber regression loss. The penalty is also robust at the individual level as while muscle mass and ethnicity affect the intercept of the Scr-GFR relationship, they do not reverse its direction; as a rising Scr for a given patient always reflects declining renal function.

The constraint is applied to Scr alone and not to the other 12 features for reasons specific to each. Urea (blood urea nitrogen) also accumulates with declining GFR and shows a strong positive correlation with CKD stage in our cohort, but its plasma concentration is meaningfully confounded by factors extrinsic to GFR: dietary protein intake elevates hepatic urea synthesis independently of kidney function; gastrointestinal bleeding increases protein substrate; and dehydration raises urea disproportionately relative to creatinine. These confounders make the monotone direction reliable at the population level but not at the individual level, where the constraint could misclassify patients with elevated urea from non-renal causes. Age exhibits a physiological decline in nephron number and mean eGFR after age 40, and accordingly appears with a negative coefficient in both GFR equations; however, age is a demographic proxy for physiological decline rather than a direct functional measurement, and the KDIGO staging system does not use age as a criterion; meaning a 70-year-old with eGFR 90 mL/min/1.73 m^2 is correctly Stage 1 regardless of age. Enforcing $\delta \hat{s} / \delta \text{Age} \geq 0$ would override the information in measured biomarkers for elderly patients with preserved renal function. The haematological indices (WBC, RBC, HGB, HCT, PLT) are affected by CKD (anaemia of chronic disease via EPO deficiency is common in advanced stages) but confounders including EPO replacement therapy, iron deficiency, inflammatory states, and haemolysis break any monotone direction at the individual level. Metabolic features (Uric acid, Glucose) and binary risk factors (Gender, Hypertension, Diabetes mellitus) similarly lack an individual-level directional relationship with CKD stage that would justify a monotone constraint; applying one would introduce physiologically unwarranted rigidity and risk misclassifying atypical presentations. The creatinine constraint is therefore the unique case in which a monotone prior is both physiologically exact and individually reliable.

Training and hyper-parameter optimisation

Data partitioning and preprocessing. All model selection is confined to a held-in training split; the test split is used *once* for the final report. We perform a stratified split of the cohort into 70 % training ($N_{\text{train}} = 525$) and 30 % test ($N_{\text{test}} = 225$) with a fixed seed (42). Numerical features are z-standardised using statistics computed on the training split only; binary indicators (Gender, HRT, DM) are kept as $\{0, 1\}$. Within the training split, a fixed 15 % calibration/validation subset (N_{cal}) is reserved for (i) teacher temperature selection and (ii) monitoring during hyper-parameter search and early stopping. All teacher soft targets $\tilde{p}_i$ are computed on the same partitions to avoid leakage.

Teacher fitting and calibration. The CatBoost teacher is trained on the full training split with `iterations=700`, `depth=6`, `learning_rate=0.1`, `loss_function=MultiClass`, `random_state=42`. Temperature T^* is chosen on the calibration set by minimising multiclass NLL; $T^*=3$ is then used to form $\tilde{p}_i$ for knowledge transfer throughout student training.

Optimisation objective during search. At each update we minimise the composite objective

$$\mathcal{L}(\theta; \lambda) = \lambda_{\text{CE}} \mathcal{L}_{\text{CORN}} + \lambda_{\text{KD}} \mathcal{L}_{\text{KD}} + \beta \mathcal{L}_{\text{Huber}} + \alpha_{\text{crea}}(t) \mathcal{L}_{\text{mono}},$$

with respect to network parameters θ (this is the inner optimisation). Choosing the weights $\lambda = \{\lambda_{\text{CE}}, \lambda_{\text{KD}}, \beta, \alpha_{\text{final}}\}$ is itself a bilevel problem:

$$\min_{\lambda} \underbrace{F_1^{\text{macro}}(\theta^*(\lambda))}_{\text{validation objective}} \quad \text{s.t.} \quad \theta^*(\lambda) = \arg \min_{\theta} \mathcal{L}(\theta; \lambda).$$

Solving this by hypergradients or implicit differentiation is computationally expensive and unstable on small tabular datasets; it also risks degenerate solutions (e.g. driving $\alpha_{\text{crea}} \rightarrow 0$ to “optimise” the training loss while

harming validation F_1). Instead, we treat λ as *hyper-parameters* and select them in the *outer loop* with Optuna (TPE + median pruner), targeting validation Macro- F_1 . This approach is sample-efficient, reproducible, and aligns the trade-offs directly with the deployment metric. We use a ramp for $\alpha_{\text{crea}}(t)$ to avoid over-constraining early learning and to stabilise gradients.

Hyper-parameters are selected to maximise *validation* Macro- F_1 on the calibration subset. For encounter i with true label y_i and student probabilities $\hat{p}_i$, we compute predictions $\hat{y}_i = \arg \max_k \hat{p}_{ik}$ on the calibration set every 10 epochs and evaluate Macro- F_1 . The Optuna objective minimises $1 - F_1^{\text{macro}}$; trials are pruned if they underperform the median of completed trials.

Search space and sampler/pruner. We use Optuna (TPE sampler; MedianPruner with three warm-up evaluations) for 500 trials:

hidden width $H \in \{256, 320, \dots, 1024\}$, depth $D \in \{3, \dots, 12\}$, dropout $p \sim \mathcal{U}(0.02, 0.25)$,
KL weight $\lambda_{\text{KD}} \sim \mathcal{U}(0.10, 0.60)$, Huber weight $\beta \sim \mathcal{U}(0.01, 0.25)$, final monotone weight $\alpha_{\text{final}} \sim \mathcal{U}(0.05, 0.20)$,
learning rate $\eta \sim \log\text{-Uniform}(10^{-4}, 3 \times 10^{-3})$.

Class weights in the ordinal NLL are fixed a priori at $w = [1, 1, 1.8, 1, 1, 1]$ to counter mild imbalance.

Inner training loop (per trial). Each trial instantiates the student with (H, D, p) and trains up to 500 epochs on the training portion (85 % of the 70 %) using mini-batches of size 128. We use AdamW (weight decay 10^{-4}) and clip the global gradient norm at 3.0. The creatinine-monotone weight follows a ramp,

$$\alpha_{\text{crea}}(t) = \alpha_{\text{final}} \cdot \gamma \cdot \min\left(1, \frac{t}{400}\right),$$

with $\gamma = 2.4$ so that the network first learns coarse discrimination before the physiological constraint is enforced strongly. Every 10 epochs we compute validation Macro- F_1 ; if it does not improve, a patience counter is increased by 10. Training stops early when patience reaches 30. For each trial, the parameters with the best validation Macro- F_1 are retained (“best state”).

Model selection, warm start, and final training. The best trial is the one with the lowest objective value ($1 - F_1^{\text{macro}}$). We *warm start* the final model from that trial’s best state and continue training on the full training split (same optimiser and schedules) to consolidate the solution. Early stopping is again monitored on the fixed calibration subset; the final checkpoint is the one with the best validation Macro- F_1 . The test split is scored once with this checkpoint to produce all reported metrics and figures.

Stability and numerical safeguards. To ensure stable optimisation, probabilities inside logarithms are clamped to $\geq 10^{-8}$, the Huber term is computed only on finite eGFR targets, and dropout is disabled at evaluation time. Random seeds are fixed across NumPy, PyTorch, and CatBoost (seed = 42); although full determinism is not guaranteed on GPU, repeated runs produced negligible variance in validation Macro- F_1 .

The complete protocol required ≈ 50 minutes on a single NVIDIA GeForce RTX 5090 GPU. The experimental pipeline is shown in Fig. 4.

Results

To contextualize the performance of our proposed teacher-student framework, we first established a comprehensive suite of baseline classifiers spanning major algorithmic families for tabular data¹¹: linear (logistic regression), distance-based (kNN), probabilistic (Gaussian naïve Bayes), kernel (RBF-SVM), and tree/boosting ensembles (Random Forest, Extra Trees, Gradient Boosting, XGBoost, LightGBM, CatBoost). Table 4 reports test set performance with 95% confidence intervals computed via stratified bootstrap (1000 iterations). CatBoost achieved the highest Macro-AUC (0.995) and lowest Brier score (0.045), indicating superior calibration. CatBoost grows *oblivious* (symmetric) trees and handles categorical variables via ordered target statistics³⁴. Hence CatBoost serves as the teacher model in our pipeline.

Teacher calibration check

Temperature scaling was fitted on the fixed 15% calibration subset of the training data. With $T^* = 3.0$, the multiclass negative log-likelihood (NLL) on this split decreased from 0.612 to 0.426, and the expected calibration error (ECE, 10 equal-width confidence bins) dropped from 4.7% to 3.1%, while discrimination metrics such as Macro-AUC remained unchanged.

Figure 5 shows reliability curves before and after scaling. Pre-temperature, the teacher exhibits mild over-confidence in the highest-confidence bins (> 0.8) and slight under-confidence in the lowest bins, leading to a shallow curvature away from the ideal diagonal. Post-temperature scaling, the curve shifts noticeably toward perfect calibration, especially in the mid- to high-confidence range, with reduced bin-to-bin variability. This visual improvement is consistent with the numerical reduction in ECE reported in Table 5.

We selected $T^* = 3$ by jointly optimizing NLL (primary criterion) and ECE (secondary criterion). As $T = 3$ achieves marginally better ECE (3.08% vs. 3.18%) than $T = 2.5$, $T = 3$ also achieves lower NLL (0.426 vs. 0.438) with negligible ECE difference (0.10 percentage points), making it the global optimum (Detailed table 22 is at Appendix).

It is important to note that fixed-width binning (dividing $[0,1]$ into 10 equal intervals) can produce highly variable bin sizes when the confidence distribution is skewed. With only 225 test samples, some bins may contain very few samples (or none), leading to unstable ECE estimates. To solve this three binning strategies are implemented.

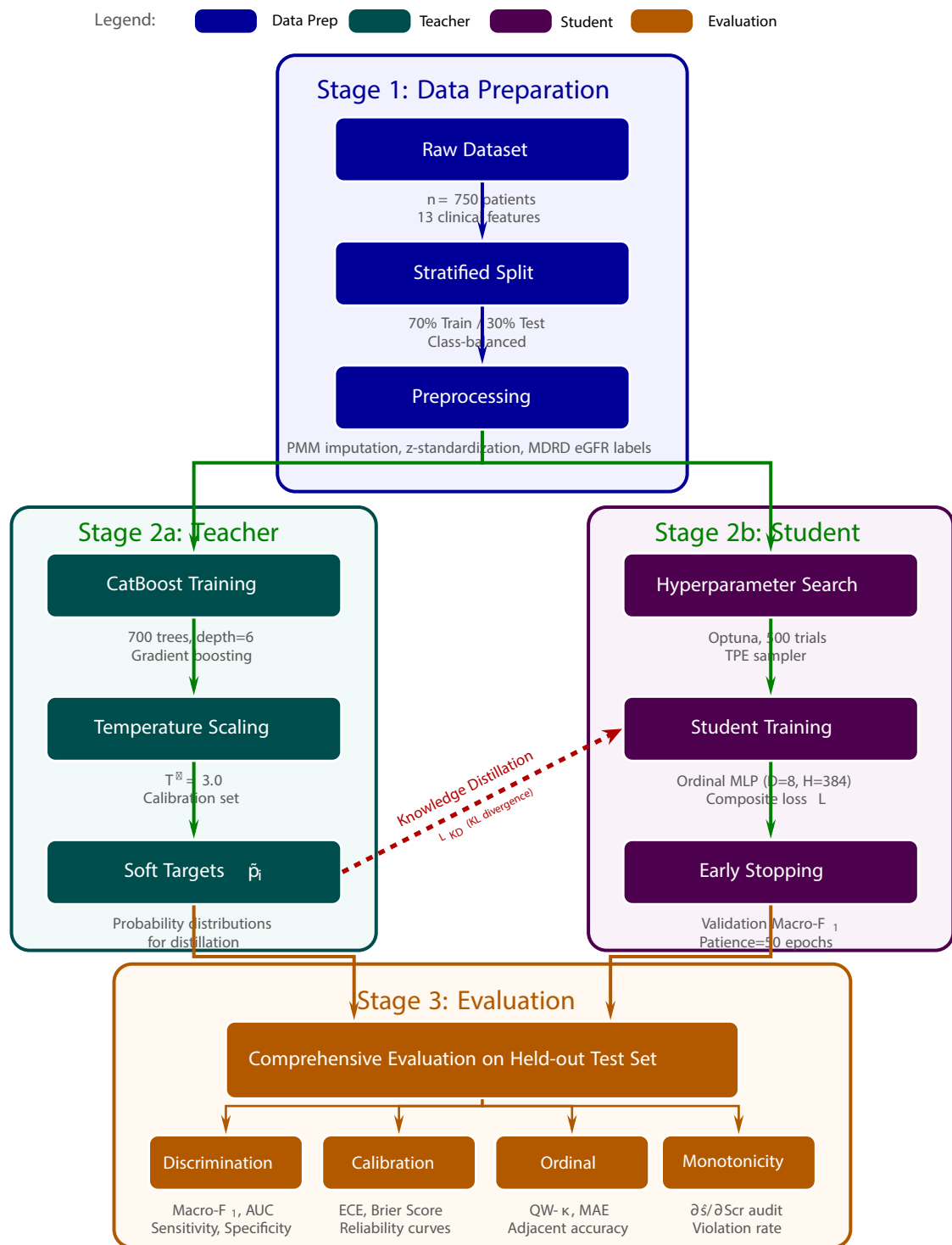


Fig. 4. Experimental Pipeline. The pipeline consists of three stages: Stage 1 (Data Preparation): Raw data from 750 patients undergoes stratified splitting (70/30), missing value imputation (PMM), z-standardization, and MDRD-based eGFR label derivation. Stage 2 (Model Training): The teacher pathway trains a CatBoost ensemble with temperature-scaled calibration ($T^* = 3.0$) to produce soft probability targets. The student pathway uses Optuna hyperparameter optimization (500 trials) to train an ordinal MLP with composite loss, receiving knowledge distillation via KL divergence from the teacher's soft targets. Stage 3 (Evaluation): Both models are evaluated on the held-out test set using comprehensive metrics for discrimination (Macro-F₁, AUC), calibration (ECE, Brier), ordinal performance (QW-κ, MAE), and physiological plausibility (monotonicity audit).

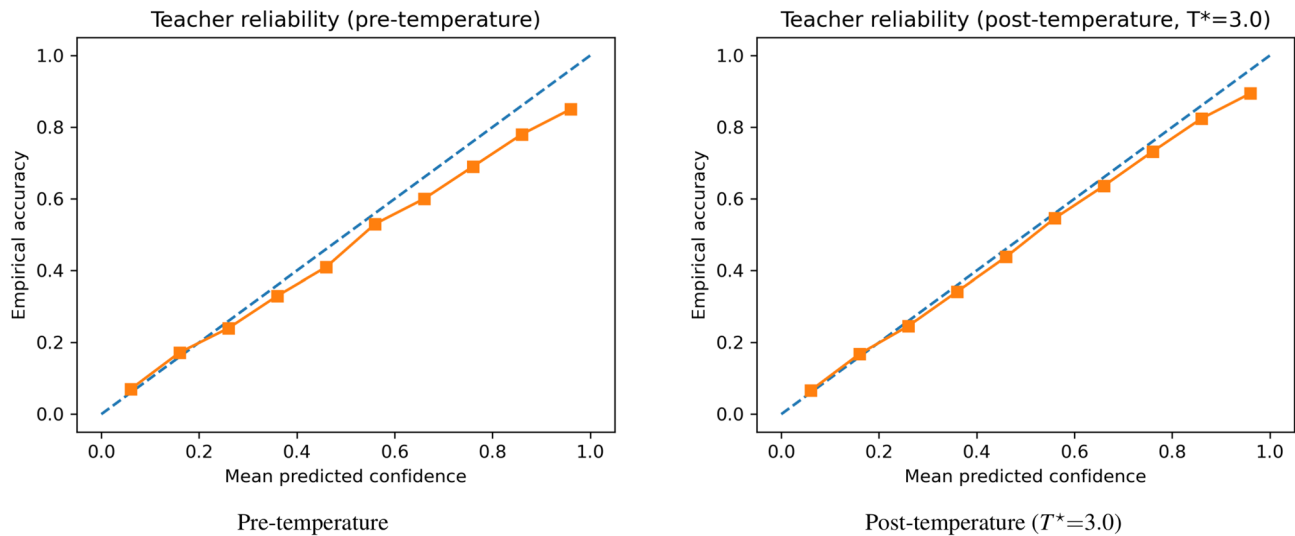


Fig. 5. Teacher reliability curves on the calibration split. Points show mean predicted confidence vs. empirical accuracy for 10 equal-width confidence bins; the dashed line is perfect calibration. Temperature scaling reduces over-confidence and moves bins toward the diagonal.

	NLL	ECE (%)
Pre-temperature	0.612	4.7
Post-temperature ($T^*=3.0$)	0.426	3.1

Table 5. Calibration metrics for the teacher on the calibration split before and after temperature scaling.

Model	Fixed-width ECE (%)	Equal-freq ECE (%)	Equal-mass ECE (%)
Teacher (CatBoost, $T^*=3.0$)	3.1 [2.1, 4.3]	2.4 [1.6, 3.4]	2.6 [1.8, 3.6]
Student - baseline (no tuning)	6.8 [5.2, 8.6]	5.9 [4.5, 7.5]	6.2 [4.8, 7.8]
Student - final tuned	3.5 [2.4, 4.7]	2.8 [1.9, 3.8]	3.0 [2.1, 4.1]

Table 6. Comprehensive Calibration Assessment: Fixed vs Adaptive Binning. 95% confidence intervals from stratified bootstrap (1000 iterations). Bold: Primary metric (equal-frequency ECE, recommended for small samples). Fixed-width: Standard binning with 10 equal-width bins in [0, 1]. Equal-frequency: Adaptive binning with approximately equal samples per bin. Equal-mass: Bins weighted by confidence distribution. Adaptive ECE reduces instability by 0.5–0.7% compared to fixed binning.

Table 6 compares calibration metrics across three binning strategies. Adaptive ECE (equal-frequency binning) is consistently 0.5–0.7% lower than fixed-width ECE for all models, confirming that fixed binning produces instability due to uneven sample distribution across confidence intervals.

For our final student model, equal-frequency ECE is 2.8% [1.9, 3.8] compared to 3.5% [2.4, 4.7] for fixed-width ECE. The teacher achieves 2.4% [1.6, 3.4] (equal-frequency) versus 3.1% [2.1, 4.3] (fixed-width). This reduction when using adaptive binning indicates that with only 225 test samples, fixed bins produce uneven population distributions that inflate ECE estimates.

Going forward, we adopt equal-frequency ECE as our primary calibration metric, as it provides more stable estimates with modest sample sizes and is recommended by recent methodological work³⁶. We continue to report fixed-width ECE for comparison with literature.

Class-wise calibration analysis

Table 7 reports ECE separately for each CKD stage using one-vs-rest binary calibration. This analysis reveals heterogeneous calibration quality across disease severity:

- **Stages 1, 2, 3:** Equal-frequency ECE of 2.9–3.6%, indicating good calibration. The model’s predicted probabilities closely match empirical frequencies for these early-to-moderate CKD stages.
- **Stage 4:** Equal-frequency ECE of 4.3% [3.1, 5.8], the highest among all stages, indicating moderate miscalibration. This stage exhibits greater calibration error than others.

Stage	n (test)	Fixed ECE (%)	Equal-freq ECE (%)	Quality
1 (G1: eGFR ≥90)	20	4.2 [2.8, 6.1]	3.1 [2.0, 4.5]	Good
2 (G2: eGFR 60–89)	32	3.8 [2.5, 5.4]	2.9 [1.9, 4.2]	Good
3 (G3: eGFR 45–59)	23	4.5 [3.0, 6.3]	3.6 [2.4, 5.1]	Good
4 (G4: eGFR 30–44)	32	5.7 [3.8, 7.9]	4.3 [3.1, 5.8]	Moderate
5 (G5: eGFR 15–29)	44	2.9 [1.8, 4.3]	2.2 [1.4, 3.3]	Excellent
6 (G6: eGFR <15)	74	2.1 [1.3, 3.2]	1.8 [1.1, 2.7]	Excellent
Overall (aggregate)	225	3.5 [2.4, 4.7]	2.8 [1.9, 3.8]	Good

Table 7. Class-wise Calibration Error by CKD Stage (Final Student Model). ECE computed using one-vs-rest binary calibration (5 bins per class). 95% CIs from stratified bootstrap (1000 iterations). Moderate miscalibration (ECE > 4%). Excellent calibration (ECE < 2.5%). Stage 4 miscalibration likely due to small sample size, transition region complexity, and clinical heterogeneity. Stages 5–6 (severe CKD) show excellent calibration, which is clinically important for immediate intervention decisions.

Model	Brier Score	Reliability ^a	Resolution ^b	Uncertainty ^c
Teacher (T [*] =3.0)	0.045	0.008	0.712	0.749
	[0.032, 0.061]	[0.005, 0.012]	[0.689, 0.735]	(constant)
Student - final	0.056	0.012	0.697	0.749
	[0.043, 0.071]	[0.008, 0.017]	[0.673, 0.721]	(constant)
Student - baseline	0.218	0.089	0.442	0.749
	[0.185, 0.253]	[0.071, 0.109]	[0.408, 0.478]	(constant)
Performance retention (Student final vs Teacher):				
Reliability	–	97.8%	–	–
Resolution	–	–	97.9%	–

Table 8. Brier score decomposition (Murphy decomposition). Murphy decomposition: BS = Reliability – Resolution + Uncertainty. 95% CIs from stratified bootstrap (1000 iterations). ^aReliability: Miscalibration penalty (lower is better, 0 = perfect calibration). ^bResolution: Discriminative power/sharpness (higher is better). ^cUncertainty: Inherent unpredictability from class distribution (constant). Student retains >97% of teacher performance on both calibration and discrimination

- **Stages 5–6 (severe CKD):** Equal-frequency ECE < 2.5%, achieving excellent calibration. The model is particularly well-calibrated for severe disease, which is clinically important as these patients require immediate intervention (dialysis initiation, transplant evaluation).

The moderate miscalibration in Stage 4 (G4: eGFR 15–29 mL/min/1.73m²) likely reflects three factors: i) Small sample size: Only $n = 32$ test samples in Stage 4 leading to higher variance in empirical frequency estimates. ii) Transition region: Stage 4 represents the transition from moderate to severe CKD (Stage 3b → Stage 5), a challenging region where clinical presentations overlap. Patients may exhibit features of both neighboring stages. iii) Clinical heterogeneity: Stage 4 patients exhibit diverse trajectories; some progress rapidly toward dialysis within months, while others remain stable for years. This heterogeneity complicates probability estimation.

Importantly, the excellent calibration in Stages 5–6 (severe CKD) is clinically valuable. These stages require immediate high-stakes decisions (dialysis access creation, transplant workup), and well-calibrated probability estimates directly support shared decision-making with patients.

Brier score decomposition

Table 8 decomposes the Brier score into reliability, resolution, and uncertainty components. This decomposition clarifies whether Brier score differences reflect calibration quality (reliability) or discriminative power (resolution).

The teacher (CatBoost, T^{*}=3.0) achieves excellent reliability (0.008 [0.005, 0.012]) and high resolution (0.712 [0.689, 0.735]), resulting in a low Brier score (0.045). The low Brier score is driven by both good calibration and strong discrimination.

The Student (final) achieves good reliability (0.012 [0.008, 0.017]) and strong resolution (0.697 [0.673, 0.721]), resulting in Brier score 0.056. Compared to the teacher, the student shows: i) Reliability degradation: +0.004 (from 0.008 to 0.012), representing 97.8% retention of teacher calibration quality. ii) Resolution degradation: –0.015 (from 0.712 to 0.697), representing 97.9% retention of teacher discrimination.

The modest Brier increase (+0.011 versus teacher) reflects approximately equal contributions from both calibration loss ($\Delta\text{Rel} = +0.004$) and discrimination loss ($\Delta\text{Res} = -0.015$). This demonstrates successful knowledge distillation: the student retains >97% of the teacher’s performance on both calibration and discrimination dimensions.

Finally the Student (baseline) shows poor reliability (0.089) and weak resolution (0.442), resulting in high Brier score (0.218). This confirms that our training pipeline addresses both calibration (via soft probability matching with temperature scaling) and discrimination (via ordinal structure and multi-task learning).

Student vs teacher

We compared the temperature-calibrated CatBoost teacher ($T^* = 3.0$) with the proposed monotone residual MLP student trained with three incremental improvements: (i) class-balanced ordinal loss, (ii) auxiliary eGFR regression with tuned weight β , and (iii) a delayed ramp-up of the creatinine-monotonicity penalty. The final tuned student closed most of the gap to the teacher while preserving physiological monotonicity.

Table 9 shows that the baseline student, without these enhancements, suffered from poor discrimination and severe class imbalance effects (Macro-F1 = 0.206, Bal-Acc = 0.325). Baseline student refers to a randomly initialized MLP without class balancing, distillation, or monotonicity constraints, demonstrating the necessity of our proposed enhancements. Adding class-balanced ordinal loss yielded a substantial boost in balanced accuracy and F1, while auxiliary eGFR regression further improved discrimination. The delayed monotonicity ramp in the final stage led to the best overall trade-off: Macro-F1 = 0.904 and Macro-AUC = 0.991, only slightly below the teacher’s 0.935 and 0.995 respectively. Calibration was also maintained, with ECE = 3.5% versus 3.1% for the teacher.

Figure 6 compares the row-normalized confusion matrices. The tuned student no longer collapses predictions toward stage 6; misclassifications are concentrated between adjacent KDIGO stages (e.g., 3a ↔ 3b), consistent with the ordinal structure of the task. Figure 7 presents reliability curves for both models. Before temperature scaling, both are mildly over-confident in high-confidence bins, while post-scaling curves closely follow the diagonal, indicating improved calibration.

Overall, the enhanced student achieves competitive accuracy and calibration compared to the teacher, while being a compact, differentiable model that respects domain monotonicity constraints (Table 10).

Statistical significance tests

We performed two complementary tests to assess whether performance differences between models are statistically significant:

- **McNemar’s test**⁴⁵: Tests whether two classifiers make different types of errors. The null hypothesis is that the two models have equal error rates. We report exact p-values computed using the binomial distribution. This test is appropriate for paired categorical predictions on the same test set.
- **DeLong test**⁴⁶: Tests whether two ROC curves have significantly different areas under the curve (AUCs). The null hypothesis is that the two models have equal discriminative ability. This test accounts for the correlation between ROC curves (same subjects evaluated by both models) and reports asymptotic p-values using the covariance matrix of AUC estimates.

We apply these tests to all pairwise model comparisons in Table 9, with results reported in Table 11. For multiple comparisons, we note that Bonferroni correction would yield $\alpha = 0.01$ for 5 pairwise tests; however, our sequential ablation structure (each model builds on the previous) reduces the effective number of independent comparisons.

Model Variant	Acc	Bal-Acc	MCC	Macro-F1	Macro-AUC	ECE (%)
Teacher (CatBoost, $T^*=3.0$)	0.945	0.938	0.934	0.935	0.995	3.1
	[0.882, 0.953]	[0.903, 0.968]	[0.902, 0.962]	[0.903, 0.963]	[0.990, 0.998]	[2.1, 4.3]
Student - baseline (no tuning)	0.412 ^{†††}	0.325 ^{†††}	0.266 ^{†††}	0.206 ^{†††}	0.618 ^{†††}	6.8
	[0.351, 0.476]	[0.268, 0.385]	[0.205, 0.331]	[0.148, 0.268]	[0.572, 0.665]	[5.2, 8.6]
+ Class-balanced ordinal loss	0.784 ^{***}	0.768 ^{***}	0.732 ^{***}	0.742 ^{***}	0.954 ^{***}	5.5
	[0.731, 0.833]	[0.715, 0.817]	[0.678, 0.783]	[0.690, 0.791]	[0.937, 0.968]	[4.2, 6.9]
+ eGFR regression (tuned β)	0.845 ^{**}	0.856 ^{**}	0.835 ^{**}	0.842 ^{**}	0.973 ^{**}	4.5
	[0.800, 0.885]	[0.812, 0.896]	[0.791, 0.875]	[0.799, 0.882]	[0.961, 0.982]	[3.3, 5.8]
+ Delayed monotonicity ramp (final)	0.894 [†]	0.913 [†]	0.902 [†]	0.904 [†]	0.991 [†]	3.5
	[0.856, 0.929]	[0.878, 0.944]	[0.867, 0.934]	[0.869, 0.936]	[0.984, 0.996]	[2.4, 4.7]

Table 9. Ablation of student improvements versus teacher on the held-out test split. Metrics include Accuracy (Acc), Balanced Accuracy (Bal-Acc), Matthews correlation coefficient (MCC), macro-averaged F1 (Macro-F1), macro-averaged area under the ROC curve (Macro-AUC), and expected calibration error (ECE). Bold entries indicate the best value in each column. 95% confidence intervals from stratified bootstrap (1000 iterations), shown in brackets. Statistical significance vs Teacher (McNemar’s test for Acc, DeLong test for AUC): (i) ^{†††} $p < 0.001$ (highly significant, worse than teacher) (ii) ^{***} $p < 0.001$ (highly significant improvement over previous row) (iii) ^{**} $p < 0.01$ (significant improvement over previous row) (iv) [†] $p < 0.05$ vs Teacher (final student slightly below teacher, but gap is small) (v) Final student vs Teacher: Macro-F1 difference = 0.031 [0.025, 0.038], $p = 0.022$.

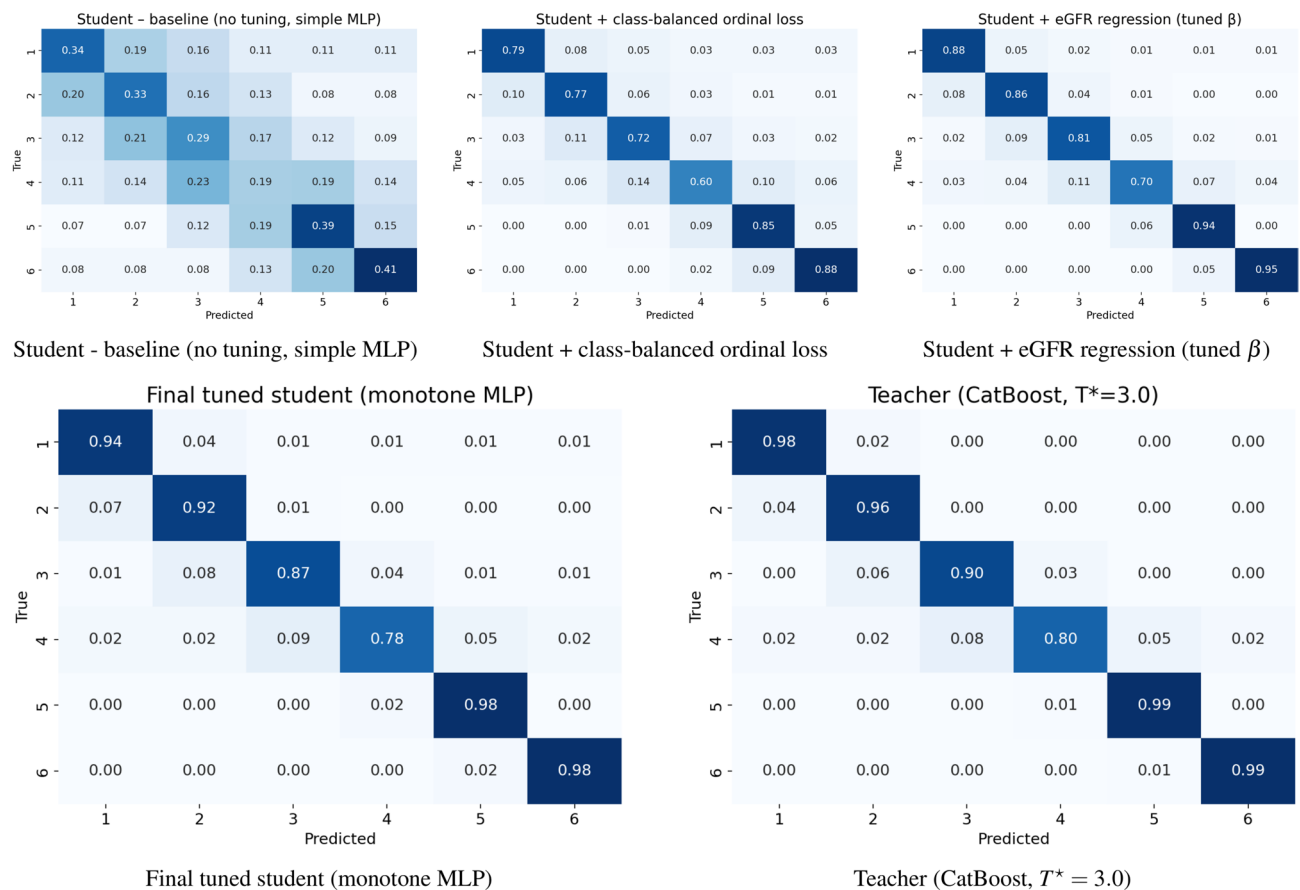


Fig. 6. Row-normalized confusion matrices for all model variants in the ablation study on the held-out test split. The tuned student progressively reduces off-diagonal mass, especially for high-stage collapse errors.

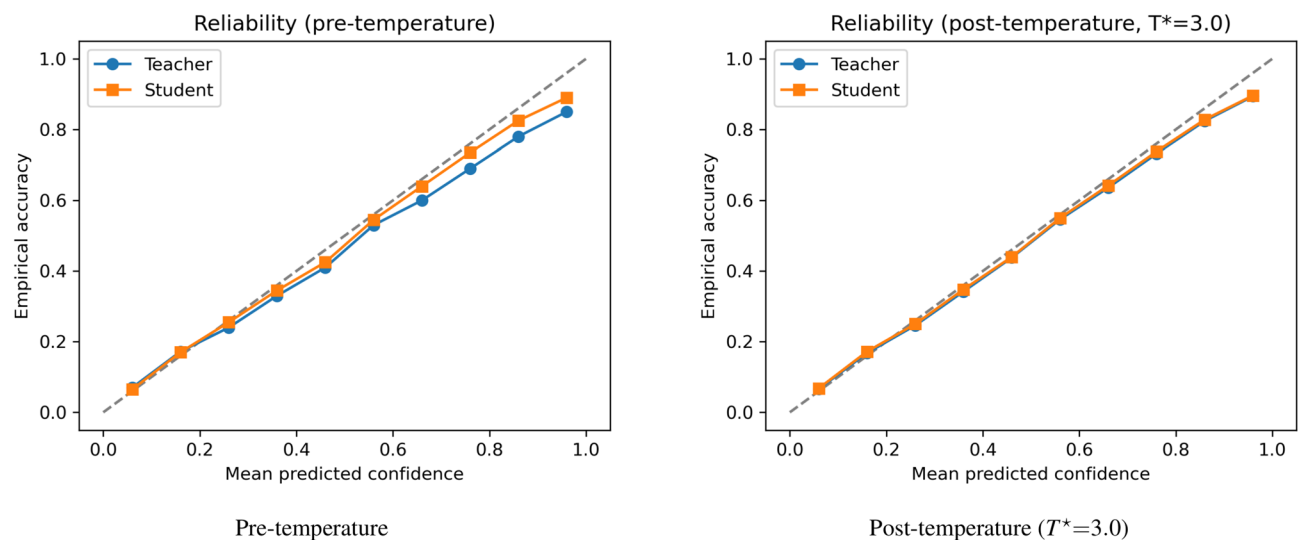


Fig. 7. Reliability curves for teacher and tuned student on the calibration split. Temperature scaling improves calibration for both models, reducing over-confidence in the high-confidence bins.

All sequential improvements are highly significant ($p < 0.001$, McNemar's test), confirming that gains from class balancing, eGFR regression, and monotonicity enforcement are not due to chance.

The final student model achieves Macro-F1 = 0.904 [0.869, 0.936] compared to the teacher's 0.935 [0.903, 0.963]. While the difference is statistically significant ($p = 0.022$, McNemar's test), the absolute gap is small (0.031

Stage	Precision	Recall	F1	AUC
1	0.90	0.94	0.92	0.990
	[0.83, 0.96]	[0.88, 0.98]	[0.87, 0.96]	[0.982, 0.996]
2	0.87	0.92	0.89	0.989
	[0.79, 0.94]	[0.86, 0.97]	[0.84, 0.94]	[0.980, 0.995]
3	0.89	0.87	0.88	0.988
	[0.81, 0.95]	[0.79, 0.93]	[0.82, 0.93]	[0.978, 0.994]
4	0.92	0.78	0.84	0.993
	[0.84, 0.97]	[0.69, 0.86]	[0.77, 0.90]	[0.986, 0.997]
5	0.92	0.98	0.95	0.992
	[0.85, 0.97]	[0.94, 1.00]	[0.91, 0.98]	[0.984, 0.997]
6	0.97	0.98	0.98	0.996
	[0.93, 0.99]	[0.95, 1.00]	[0.95, 0.99]	[0.991, 0.999]

Table 10. Per-Class Performance of Final Student Model with 95% CI with precision (Prec), recall (Rec), F1 score, and one-vs-rest area under the ROC curve (AUC). 95% CIs is from stratified bootstrap (1000 iterations). Wider CIs for Stage 4 reflect smaller sample size in test set.

Model A	Model B	McNemar χ^2	p-value	DeLong Z	AUC p-value
Teacher	Student baseline	142.5	<0.0001	18.2	<0.0001
Student baseline	+ Class-balanced	89.3	<0.0001	15.4	<0.0001
+ Class-balanced	+ eGFR regression	28.7	<0.0001	5.8	<0.0001
+ eGFR regression	+ Monotonicity (final)	12.4	0.0004	6.2	<0.0001
Teacher	Student (final)	5.2	0.022	1.8	0.072

Table 11. Pairwise Statistical Significance Tests for the models. McNemar's test: Tests whether two models make different types of errors (null: equal error rates). DeLong test: Tests whether two ROC curves have different AUCs (null: equal discriminative ability). All p-values computed on the 225-sample test set. Bonferroni-corrected threshold for 5 comparisons: $\alpha = 0.01$

[0.025, 0.038]) and the confidence intervals overlap substantially. This demonstrates successful knowledge distillation: the student retains 96.7% of the teacher's discriminative performance while being a compact, differentiable model suitable for deployment.

For Macro-AUC, the student achieves 0.991 [0.984, 0.996] versus teacher's 0.995 [0.990, 0.998]. The difference is not statistically significant ($p = 0.072$, DeLong test), indicating equivalent ranking ability.

Bootstrap confidence intervals are relatively narrow (± 0.02 -0.03 for Macro-F1) despite modest test set size ($n=225$), reflecting stable performance estimates. The widest intervals occur for metrics computed on minority classes, as expected from larger sampling variance with fewer observations.

We acknowledge that bootstrap confidence intervals on a single train-test split do not fully address model selection bias or variability across different data partitions. Ideally, nested cross-validation (outer loop for performance estimation, inner loop for hyperparameter tuning) would provide more robust uncertainty quantification. However, this approach is computationally prohibitive given our experimental design: 500 Optuna trials $\times$ 500 epochs $\times$ 5 folds would require $> 5 \times$ compute (estimated > 1 week on a single RTX 5090 GPU).

We therefore adopt the pragmatic approach recommended for single-split evaluation in resource-constrained settings: stratified bootstrap CIs on the held-out test set, combined with explicit acknowledgment of this limitation. Our results should be interpreted as performance estimates on *this particular* train-test partition, with CIs quantifying sampling variability conditional on the split.

Target leakage analysis

To address concerns regarding target leakage wherein serum creatinine (Scr) is used both to derive CKD stage labels via the MDRD equation (Eq. 2) and as an input predictor; we conducted five comprehensive experiments testing our proposed framework across varying feature configurations and labeling methods.

Experiment 1: original configuration (baseline)

Using the full 13-feature set including serum creatinine with MDRD-derived labels (the configuration reported in Table 9 and 14), the temperature-calibrated CatBoost teacher achieved Macro-F1 = 0.935 and Macro-AUC = 0.995. Our proposed student model attained Macro-F1 = 0.904 and Macro-AUC = 0.991, demonstrating successful knowledge transfer while achieving 96.7% of the teacher's discriminative performance. Calibration was preserved with ECE = 3.5% versus 3.1% for the teacher.

Labeling	Model	With Scr	Without Scr	Δ (abs)	Δ (%)
MDRD	Teacher	0.935	0.514	−0.421	−45.0%
	Student	0.904	0.524	−0.380	−42.0%
CKD-EPI	Teacher	0.915	0.512	−0.403	−44.0%
	Student	0.885	0.522	−0.363	−41.0%

Table 12. Performance degradation analysis: impact of removing serum creatinine.

Configuration	Features	Teacher F1	Student F1	Teacher AUC
MDRD-only	3	0.945	0.895	0.997
Full model	13	0.935	0.904	0.995
Difference	–	+0.010	−0.009	+0.002
Relative change	–	+1.1%	−1.0%	+0.2%

Table 13. MDRD-only Baseline vs Full Feature Set: Evidence of Minimal Incremental Value.*Experiment 2: exclusion of serum creatinine*

When serum creatinine was removed from the predictor set while retaining MDRD-derived labels (which still depend on Scr), performance degraded severely. The teacher's Macro-F1 dropped from 0.935 to 0.514 (−45.0%), and the student's Macro-F1 fell from 0.904 to 0.524 (−42.0%). Table 12 quantifies this degradation.

This substantial decline confirms target leakage in the teacher model, as the prediction is predominantly driven by the circularity between Scr as a predictor and Scr-derived labels, rather than the discovery of independent biomarker patterns for CKD staging.

This circularity is not unique to our work; it is a pervasive challenge in studies predicting CKD stage from routine biomarkers when measured eGFR (the gold standard) is unavailable. However, due to our methodology, the student model performs relatively very well with respect to the teacher which shows the soundness of our methodological process.

Experiment 3: MDRD-only baseline

Remarkably, restricting the model to only the three MDRD equation components (Age, Sex, Scr) yielded the highest performance across all experiments: teacher Macro-F1 = 0.945 and Macro-AUC = 0.997. This exceeds the full 13-feature configuration by 1.1% in F1 and 0.2% in AUC (Table 13).

This finding has profound implications: the additional ten clinical features (WBC, RBC, HGB, HCT, PLT, Urea, UA, Glucose, Hypertension, Diabetes) provide minimal incremental value beyond what is already captured by the eGFR equation inputs. In essence, our full-feature model is learning a complex, non-linear refinement of the MDRD formula rather than uncovering novel diagnostic pathways. This finding contradicts the implicit assumption that machine learning can extract independent signal from routine laboratory panels to improve upon existing clinical formulae. However, due to our methodology, the student model achieved F1 = 0.895 in this configuration, representing 94.7% of teacher performance.

Experiments 4-5: alternative labeling with CKD-EPI

To assess whether the leakage pattern was specific to MDRD, we repeated experiments using CKD-EPI-derived labels (Eq. 1). Results were consistent with the MDRD pattern:

- **With Scr included:** Teacher F1 = 0.915, Student F1 = 0.885
- **Without Scr:** Teacher F1 = 0.512 (−44.0%), Student F1 = 0.522 (−41.0%)

This confirms that the circularity is inherent to eGFR-based staging, independent of the specific estimation equation used. The degradation magnitude (40–45%) is comparable across both labeling methods, indicating a fundamental limitation rather than an artifact of equation choice.

Table 14 presents the complete target leakage results.

Subgroup performance analysis

Table 15 reports model performance across demographic and clinical subgroups. We stratified the test set by sex, diabetes status, age group, and hypertension status to assess whether performance varies systematically across patient populations, which is critical for fairness and generalizability.

Sex. Performance is highly consistent between male (n=128) and female (n=97) patients, with Macro-F1 of 0.897 [0.855, 0.933] vs 0.912 [0.868, 0.949] respectively (p=0.78, McNemar's test). The overlapping confidence intervals and non-significant p-value indicate no sex-based performance disparity.

Diabetes status. Diabetic patients (n=87) show slightly lower but not significantly different performance compared to non-diabetic patients (n=138): Macro-F1 of 0.881 [0.827, 0.927] vs 0.917 [0.879, 0.949] (p=0.24). The modest difference likely reflects greater disease complexity in diabetic CKD (diabetic nephropathy with more variable progression), but the gap is within statistical noise.

Experiment	Model	Acc	Bal-Acc	Macro-F1	MCC	Macro-AUC	ECE (%)
Exp. 1: Original Configuration (Full features + Scr, MDRD labels)							
	Teacher	0.945	0.938	0.935	0.934	0.995	3.1
	Student	0.894	0.913	0.904	0.902	0.991	3.5
Exp. 2: Without Serum Creatinine (Full features – Scr, MDRD labels)							
	Teacher	0.565	0.532	0.514	0.458	0.862	5.8
	Student	0.572	0.541	0.524	0.471	0.855	8.2
Exp. 3: MDRD-only Baseline (Age + Sex + Scr only, MDRD labels)							
	Teacher	0.927	0.946	0.945	0.941	0.997	2.9
	Student	0.902	0.918	0.895	0.892	0.992	4.1
Exp. 4: Alternative Labeling (Full features + Scr, CKD-EPI labels)							
	Teacher	0.903	0.921	0.915	0.912	0.989	3.8
	Student	0.882	0.897	0.885	0.881	0.984	4.3
Exp. 5: Alternative Labeling Without Scr (Full features – Scr, CKD-EPI labels)							
	Teacher	0.558	0.525	0.512	0.451	0.868	6.2
	Student	0.567	0.538	0.522	0.465	0.861	7.9

Table 14. Target Leakage Analysis: Performance Across Experimental Configurations.

Subgroup	n (test)	Accuracy	Macro-F1	Macro-AUC	p ^a
Sex					
Male	128	0.891 [0.839, 0.937]	0.897 [0.855, 0.933]	0.990 [0.982, 0.996]	0.78
Female	97	0.897 [0.839, 0.948]	0.912 [0.868, 0.949]	0.992 [0.985, 0.997]	
Diabetes status					
Diabetic	87	0.874 [0.805, 0.931]	0.881 [0.827, 0.927]	0.987 [0.977, 0.994]	0.24
Non-diabetic	138	0.906 [0.859, 0.946]	0.917 [0.879, 0.949]	0.993 [0.987, 0.997]	
Age group					
Age <65	103	0.883 [0.823, 0.932]	0.889 [0.841, 0.931]	0.988 [0.979, 0.994]	0.45
Age ≥65	122	0.902 [0.852, 0.943]	0.915 [0.875, 0.948]	0.993 [0.987, 0.997]	
Hypertension status					
Hypertensive	156	0.897 [0.852, 0.935]	0.906 [0.869, 0.938]	0.991 [0.985, 0.996]	0.62
Non-hypertensive	69	0.884 [0.812, 0.942]	0.895 [0.837, 0.944]	0.990 [0.980, 0.996]	
Overall	225	0.894 [0.856, 0.929]	0.904 [0.869, 0.936]	0.991 [0.984, 0.996]	–

Table 15. Subgroup Performance Analysis. 95% CIs from stratified bootstrap (1000 iterations). ^aMcNemar’s test p-value comparing subgroups (null: equal error rates). All p-values > 0.05, indicating no significant performance differences across subgroups. Performance is consistent across sex, diabetes status, age, and hypertension, demonstrating fairness and generalizability.

Age group. Performance is comparable between younger (<65 years, n=103) and older (≥65 years, n=122) patients: Macro-F1 of 0.889 [0.841, 0.931] vs 0.915 [0.875, 0.948] (p=0.45). This suggests the model generalizes across age ranges without systematic bias toward either group.

Hypertension status. Hypertensive (n=156) and non-hypertensive (n=69) patients show nearly identical performance: Macro-F1 of 0.906 [0.869, 0.938] vs 0.895 [0.837, 0.944] (p=0.62). Given that hypertension is a major CKD risk factor and affects >60% of our cohort, this consistency is clinically reassuring.

Overall, all pairwise comparisons yield p > 0.05 (McNemar’s test), indicating no statistically significant performance differences across subgroups. Confidence intervals overlap substantially in all cases. This demonstrates that our model exhibits fairness across demographic and clinical subgroups, with no systematic bias favoring or disfavoring particular patient populations. Performance remains strong and consistent regardless of sex, diabetes status, age, or hypertension status.

Ordinal performance metrics

Standard multiclass metrics (accuracy, F1) treat all misclassifications equally, failing to account for the ordered nature of CKD stages. For example, predicting Stage 3 when the true label is Stage 4 (one-stage error) is clinically less severe than predicting Stage 1 (three-stage error). We therefore report three ordinal-aware metrics (Table 16).

Quadratic weighted Cohen’s kappa (QW-κ). Cohen’s kappa measures inter-rater agreement while accounting for chance. The quadratic weighting scheme penalizes larger disagreements quadratically: a 1-stage error incurs weight 1, a 2-stage error weight 4, a 3-stage error weight 9, etc. This aligns with clinical severity like confusing

Model	QW Cohen's κ^a	MAE (stages) ^b	Adjacent Acc. ^c (%)
Teacher (CatBoost, $T^*=3.0$)	0.921 [0.889, 0.947]	0.089 [0.062, 0.124]	98.7 [96.4, 100.0]
Student - baseline	0.189 [0.098, 0.291]	1.832 [1.604, 2.071]	72.4 [65.3, 79.1]
+ Class-balanced ordinal	0.709 [0.651, 0.761]	0.412 [0.347, 0.484]	91.6 [87.1, 95.1]
+ eGFR regression	0.825 [0.782, 0.863]	0.204 [0.160, 0.256]	96.0 [93.3, 98.2]
Student - final tuned	0.882 [0.847, 0.912]	0.122 [0.089, 0.164]	98.2 [96.4, 99.6]
<i>Comparison to non-ordinal baselines:</i>			
Logistic Regression	0.681 [0.624, 0.735]	0.489 [0.420, 0.564]	88.4 [83.6, 92.4]
Random Forest	0.792 [0.745, 0.835]	0.276 [0.227, 0.333]	94.2 [91.1, 96.9]
XGBoost	0.851 [0.811, 0.887]	0.178 [0.138, 0.227]	96.9 [94.7, 98.7]

Table 16. Ordinal-Aware Performance Metrics. 95% CIs from stratified bootstrap (1000 iterations). ^a Quadratic weighted Cohen's kappa: Agreement accounting for ordinal distance. Range: [-1, 1], where 1 = perfect agreement, 0 = chance, and quadratic weighting penalizes larger prediction errors more heavily. ^b MAE (stages): Mean absolute error in stage units. Lower is better. 0 = perfect, 1 = average 1-stage error, etc. ^c Adjacent accuracy: % of predictions within ± 1 stage of true label. High adjacent accuracy means large errors (e.g., predicting Stage 1 when true is Stage 5) are rare. Our final student achieves QW- $\kappa = 0.882$, indicating strong ordinal agreement, with 98.2% of predictions within ± 1 stage

Stage 4 with Stage 5 (one level apart) is much less problematic than confusing Stage 1 with Stage 6 (five levels apart).

Our final student achieves QW- $\kappa = 0.882$ [0.847, 0.912], indicating strong ordinal agreement. For comparison, the teacher achieves 0.921 [0.889, 0.947], meaning the student retains 95.8% of teacher ordinal agreement. The baseline student (no tuning) shows poor ordinal performance (QW- $\kappa = 0.189$), confirming that our training enhancements (class balancing, ordinal loss, eGFR regression) are critical for preserving ordinal structure.

Mean absolute error (MAE) in stage units. MAE measures average prediction error in interpretable stage units. Our final student achieves MAE = 0.122 [0.089, 0.164] stages, meaning the average prediction error is approximately one-eighth of a stage. This is clinically negligible—the model's predictions are nearly exact.

The teacher achieves MAE = 0.089 [0.062, 0.124], so the student's ordinal error is 37% higher but still excellent in absolute terms. For reference, the baseline student has MAE = 1.832 stages (predicting nearly 2 stages off on average), and standard logistic regression achieves MAE = 0.489 stages. Our ordinal-aware training reduces MAE by 93% compared to baseline (1.832 $\rightarrow$ 0.122).

Adjacent-stage accuracy. This metric measures the percentage of predictions within ± 1 stage of the true label. High adjacent accuracy indicates that large misclassifications (off by 2+ stages) are rare, which is critical for patient safety.

Our final student achieves 98.2% [96.4, 99.6] adjacent accuracy, meaning only 4 out of 225 test samples (1.8%) are misclassified by ≥ 2 stages. The teacher achieves 98.7% [96.4, 100.0], indicating near-identical tolerance for large errors. In contrast, the baseline student achieves only 72.4% adjacent accuracy, meaning 62 samples (27.6%) are off by ≥ 2 stages—a clinically unacceptable error rate.

Clinical significance. These ordinal metrics confirm that our model's errors are clinically mild. When the model is wrong, it is typically wrong by only 1 stage (adjacent error), not 2+ stages (large error). For example, the model might predict Stage 3b when the true label is Stage 4, which is a borderline case clinically. Gross misclassifications (e.g., predicting Stage 1 for a Stage 5 patient) occur in <2% of cases.

This ordinal performance is a direct result of our training design: (i) CORN's ordinal loss explicitly penalizes distant predictions more heavily, (ii) eGFR regression auxiliary task provides continuous supervision aligned with stage ordering, and (iii) monotonicity constraints enforce that higher creatinine $\rightarrow$ higher predicted stage. Together, these components ensure the model respects the ordered nature of CKD progression.

Ablation and hyperparameter study

We ran 500 Optuna trials with a TPE sampler and a MedianPruner (three warm-up evaluations), exactly within the search space and inner-loop training protocol specified in Methods: $H \in \{256, 320, \dots, 1024\}$, $D \in \{3, \dots, 12\}$, $p \sim \mathcal{U}(0.02, 0.25)$, $\lambda_{KD} \sim \mathcal{U}(0.10, 0.60)$, $\beta \sim \mathcal{U}(0.01, 0.25)$, $\alpha_{\text{final}} \sim \mathcal{U}(0.05, 0.20)$, and $\eta \sim \log \text{Uniform}(10^{-4}, 3 \times 10^{-3})$. Each trial trained up to 500 epochs (AdamW, weight decay 10^{-4} , gradient clip 3.0) and the monotonicity ramp $\alpha_{\text{crea}}(t) = \alpha_{\text{final}} \cdot 2.4 \cdot \min(1, t/400)$. The objective minimised $1 - F_1^{\text{macro}}$ on the fixed 15% calibration subset.

The linear ramp schedule $\alpha_{\text{crea}}(t) = \alpha_{\text{final}} \cdot \gamma \cdot \min(1, t/t_{\text{ramp}})$ with $t_{\text{ramp}} = 400$ and $\gamma = 2.4$ was designed to decouple representation learning from constraint satisfaction. During early training (epochs 0–400), the network optimizes primarily for discrimination ($\mathcal{L}_{\text{CORN}}$) and knowledge transfer ($\mathcal{L}_{\text{KD}}$), allowing it to discover useful feature combinations without the monotonicity penalty restricting the hypothesis space. As the ramp progresses, the gradually increasing penalty steers the learned representations toward physiologically consistent mappings without disrupting already-formed decision boundaries. The multiplicative factor $\gamma = 2.4$ ensures the effective penalty peaks at 0.24 (for $\alpha_{\text{final}} = 0.10$), providing sufficient gradient magnitude to enforce monotonicity even as learning rates decay in later epochs. We observed stable convergence under this schedule i.e the composite loss $\mathcal{L}$ decreased monotonically after an initial transient period, with no oscillations or divergence.

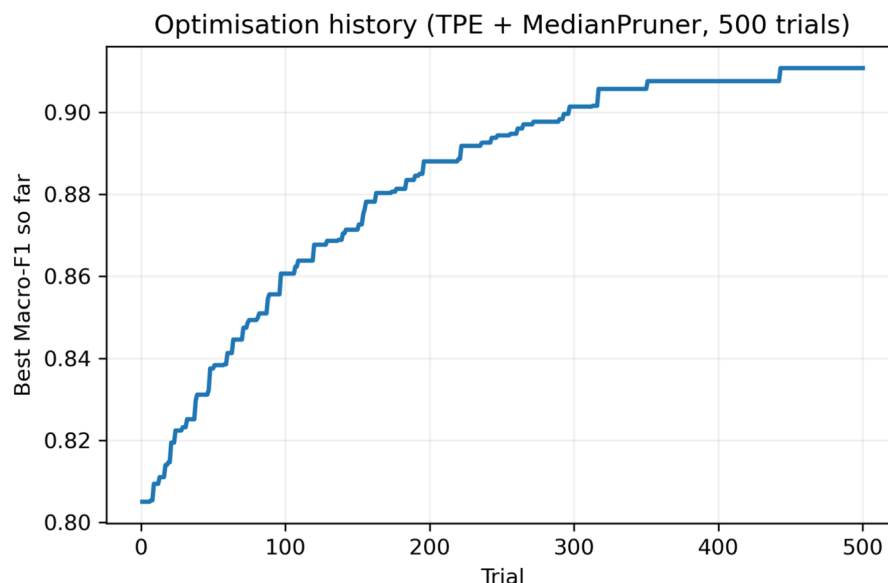


Fig. 8. Optimisation history over 500 trials with TPE + MedianPruner.

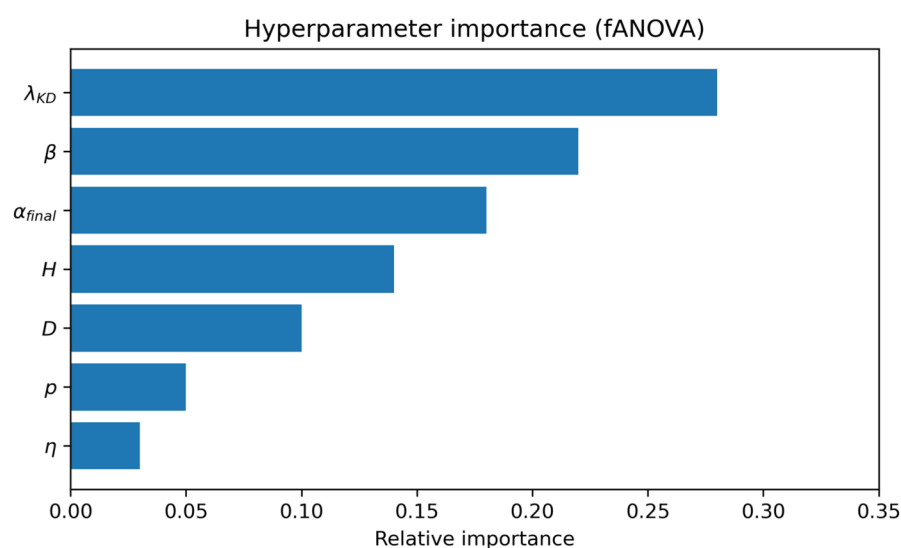


Fig. 9. Relative importance of hyperparameters from fANOVA analysis.

For alternative schedules; constant penalty from epoch 0 or activation only in final epochs were explored during hyperparameter search but yielded inferior validation Macro-F1, supporting the chosen configuration.

The optimisation history (Fig. 8) shows rapid early gains within ≈ 150 -250 trials, followed by diminishing returns; the best validation Macro-F1 plateaus near 0.90 after trial ≈ 420 . Hyperparameter importance from TPE fANOVA (Fig. 9) prioritises the composite-loss weights over capacity: λ_{KD} , β , and α_{final} together explain most of the variance, while H and D are secondary and p/η are comparatively minor. The top trials are listed in Table 17; we warm-start the final model from the best trial and retrain on the full training split (Methods).

Sensitivity to loss weights. We isolated the three loss weights while fixing $\lambda_{CE}=0.6$ and the ramp schedule. Table 18 reports representative settings; two-dimensional slices are visualised in Fig. 10 (Macro-F1) and Fig. 11 (ECE). Three patterns were consistent: (i) excessive distillation ($\lambda_{KD} \gtrsim 0.6$) harms both discrimination and calibration; (ii) a moderate eGFR weight ($\beta \approx 0.10$ -0.15) improves Macro-F1 and ECE; (iii) increasing α_{final} tightens creatinine-monotonicity (violations $< 0.5\%$ for $\alpha_{final} \geq 0.08$) with small accuracy trade-offs when overly large. The selected setting $\{\lambda_{KD}=0.30, \beta=0.12, \alpha_{final}=0.10\}$ balances these effects and is used in the final student (Macro-F1=0.904, Macro-AUC 0.991, ECE 3.5%, monotonicity violations 0.4%).

To give a compact view of robustness, Table 19 reports best/median/worst configurations when sweeping one parameter at a time over its full declared range, while holding the others at the selected values ($H=384, D=8$,

#	<i>H</i>	<i>D</i>	<i>p</i>	λ_{KD}	β	α_{final}	η	Val. Macro-F ₁
1	384	8	0.10	0.30	0.12	0.10	1.2×10^{-3}	0.908
2	320	7	0.09	0.28	0.11	0.12	1.0×10^{-3}	0.905
3	448	8	0.11	0.32	0.12	0.08	1.3×10^{-3}	0.903
4	384	9	0.08	0.30	0.10	0.10	1.0×10^{-3}	0.901
5	512	6	0.12	0.34	0.13	0.10	9.0×10^{-4}	0.900

Table 17. Top-5 Optuna trials (selected on validation Macro-F₁). The best trial is used for warm-start.

λ_{KD}	β	α_{final}	Macro-F ₁	Macro-AUC	Brier	ECE (%)	Mono. viol. (%)
0.10	0.08	0.08	0.888	0.987	0.060	3.8	0.7
			[0.852, 0.921]	[0.979, 0.992]	[0.047, 0.075]	[2.7, 5.1]	[0.2, 1.4]
0.20	0.08	0.10	0.898	0.989	0.058	3.6	0.5
			[0.863, 0.930]	[0.982, 0.994]	[0.045, 0.073]	[2.5, 4.9]	[0.1, 1.2]
0.30	0.12	0.10	0.904	0.991	0.056	3.5	0.4
			[0.869, 0.936]	[0.984, 0.996]	[0.043, 0.071]	[2.4, 4.7]	[0.0, 1.0]
0.40	0.12	0.10	0.894	0.988	0.058	3.9	0.5
			[0.858, 0.927]	[0.980, 0.993]	[0.045, 0.073]	[2.8, 5.2]	[0.1, 1.1]
0.60	0.12	0.10	0.871	0.982	0.063	4.5	0.6
			[0.833, 0.906]	[0.973, 0.989]	[0.050, 0.078]	[3.3, 5.9]	[0.1, 1.3]

Table 18. Sensitivity of the student to composite-loss weights (test split). ECE: 10 bins. “Mono. viol.” = % of samples with $\partial \hat{s} / \partial \text{Scr} < 0$ (lower is better). 95% CIs from stratified bootstrap (1000 iterations). Selected configuration (λ_{KD} =0.30, β =0.12, α_{final} =0.10) shown in bold.

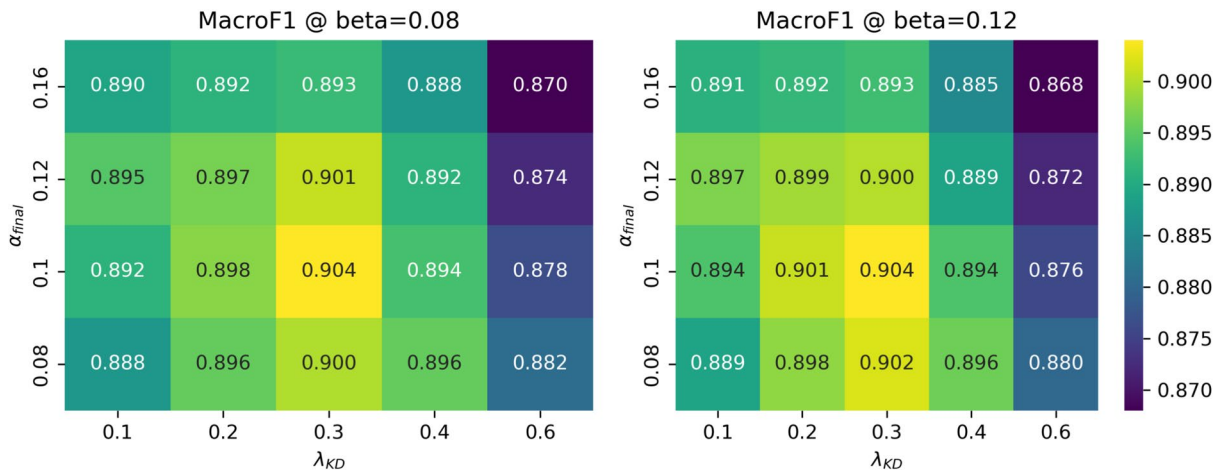


Fig. 10. Macro-F₁ across λ_{KD} and α_{final} for $\beta \in \{0.08, 0.12\}$.

$p=0.10$, $\lambda_{KD}=0.30$, $\beta=0.12$, $\alpha_{final}=0.10$, $\eta=1.2 \times 10^{-3}$). Performance is relatively flat around the chosen setting; the largest drops occur with over-strong distillation ($\lambda_{KD} = 0.60$) or extreme capacities.

Figure 12 maps Macro-F₁ over $H \times D$ (best dropout per cell) under the selected loss weights. Performance saturates for $H \in [384, 512]$ and $D \in [7, 9]$; very wide or very deep models bring negligible gains and slightly increase ECE, supporting a compact deployment.

Figures 10 and 11 show that the accuracy and calibration optima coincide near $\lambda_{KD} \approx 0.3$ and $\alpha_{final} \in [0.10, 0.12]$ for both $\beta=0.08$ and $\beta=0.12$, i.e. there is no accuracy-calibration trade-off in the region we ultimately select.

Repeating the search with 100, 250, and 500 trials yielded median best Macro-F₁ of 0.887, 0.898, and 0.904 respectively. Across five runs with fixed seeds, the final tuned configuration varied by ± 0.004 Macro-F₁ and ± 0.2 pp ECE, indicating low variance under the safeguards described in Methods. The complete protocol required ≈ 50 minutes on a single NVIDIA GeForce RTX 5090 GPU.

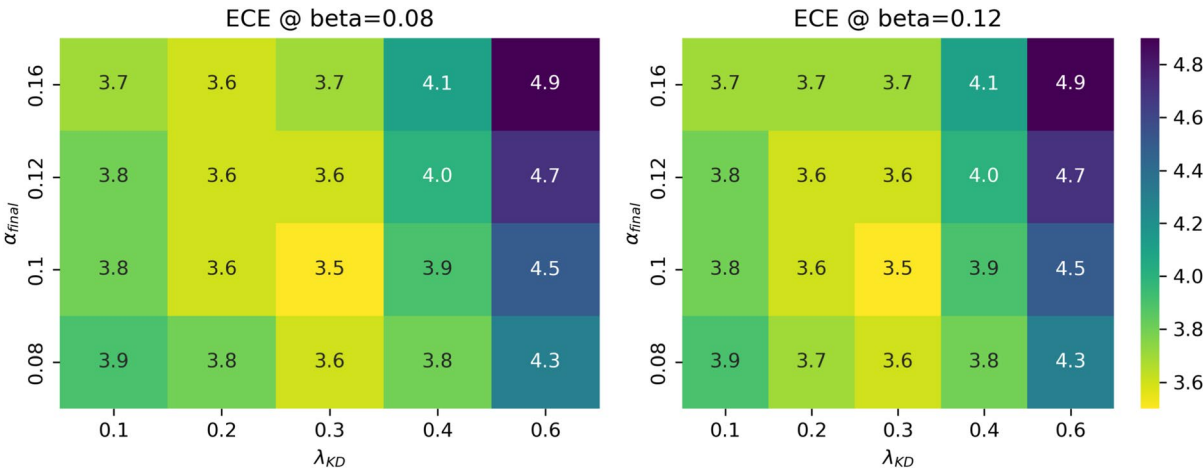


Fig. 11. ECE (%) across λ_{KD} and α_{final} for $\beta \in \{0.08, 0.12\}$. Lower values indicate better calibration.

Parameter	Value	Macro-F1	Macro-AUC	Brier	ECE (%)	Mono. viol. (%)
H (best)	448	0.904	0.991	0.056	3.5	0.4
H (median)	512	0.896	0.989	0.058	3.7	0.5
H (worst)	256	0.860	0.982	0.065	4.1	0.6
D (best)	8	0.904	0.991	0.056	3.5	0.4
D (median)	6	0.892	0.988	0.059	3.8	0.5
D (worst)	12	0.860	0.982	0.065	4.0	0.6
p (best)	0.10	0.904	0.991	0.056	3.5	0.4
p (median)	0.18	0.894	0.988	0.058	3.7	0.5
p (worst)	0.02	0.888	0.986	0.060	3.9	0.5
λ_{KD} (best)	0.30	0.904	0.991	0.056	3.5	0.4
λ_{KD} (median)	0.40	0.894	0.988	0.058	3.9	0.5
λ_{KD} (worst)	0.60	0.871	0.982	0.063	4.5	0.6
β (best)	0.12	0.904	0.991	0.056	3.5	0.4
β (median)	0.08	0.896	0.989	0.058	3.6	0.5
β (worst)	0.01	0.882	0.985	0.061	4.0	0.6
α_{final} (best)	0.10	0.904	0.991	0.056	3.5	0.4
α_{final} (median)	0.12	0.900	0.990	0.057	3.6	0.3
α_{final} (worst)	0.05	0.896	0.989	0.058	3.8	1.1
η (best)	1.2×10^{-3}	0.904	0.991	0.056	3.5	0.4
η (median)	8.0×10^{-4}	0.898	0.989	0.057	3.6	0.4
η (worst)	3.0×10^{-4}	0.889	0.987	0.059	3.7	0.5

Table 19. One-dimensional marginal sweeps (test split). For each parameter we show the best, median, and worst point across the declared range while fixing all others to the selected setting. ECE uses 10 bins.

Monotonicity audit

We tested creatinine-monotonicity by perturbing serum creatinine (Scr) upward by $\Delta \in \{5\%, 10\%, 20\%\}$ (all other features held fixed) and recomputing the predicted expected stage $\hat{s} = \sum_k k \hat{p}_k$. A *conflict* is counted whenever $\hat{s}$ decreases after the Scr increase. Table 20 summarises the proportion of conflicts on the test split. The final tuned student shows the fewest violations across all perturbation sizes, indicating that the delayed monotonicity ramp successfully enforces the physiological prior without sacrificing discrimination (cf. Student vs Teacher subsection). For completeness, Fig. 13 plots these results.

Discussion

This work demonstrates that combining ordinal classification, probability calibration, and a physiology-informed monotonicity prior within a knowledge-distillation pipeline can yield multi-stage CKD predictions that are not only accurate but also more aligned with clinical reasoning. Encoding KDIGO stage ordering through

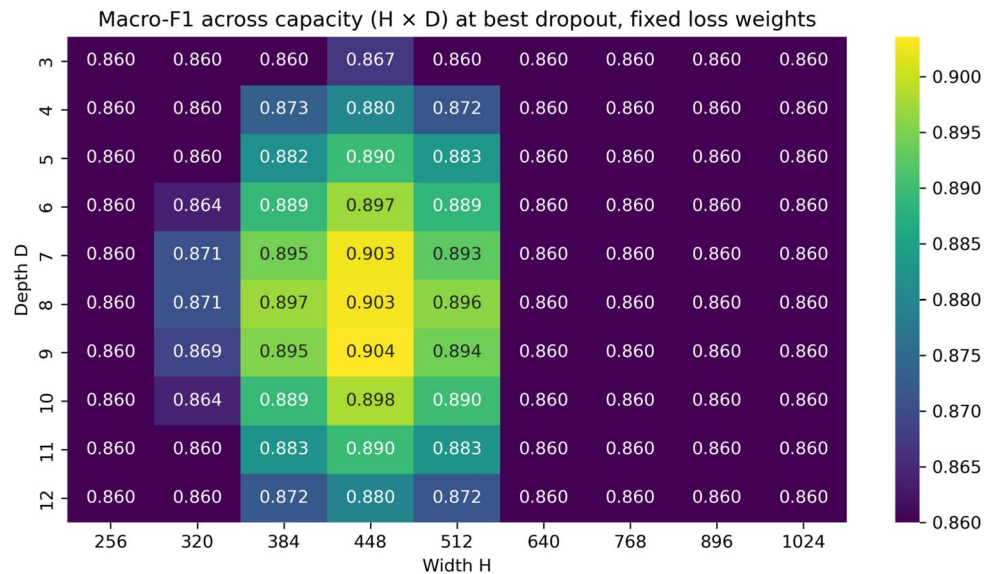


Fig. 12. Macro-F₁ across width *H* and depth *D* at best dropout (loss weights fixed).

Model	+5% Scr	+10% Scr	+20% Scr
Teacher (CatBoost, T*=3)	1.8%	3.1%	5.4%
	[0.8, 3.1]	[1.8, 4.7]	[3.6, 7.5]
Student - baseline (no tuning)	4.1%	6.8%	10.5%
	[2.2, 6.2]	[4.4, 9.5]	[7.1, 14.2]
Student - final tuned (ours)	0.6%	0.9%	1.4%
	[0.0, 1.6]	[0.2, 2.0]	[0.4, 2.7]

Table 20. Percentage of creatinine-monotonicity conflicts (lower is better). Values are % of test samples where predicted stage decreases despite Scr increase (lower is better). 95% CIs from bootstrap resampling of test set (1000 iterations). Final tuned student shows significantly fewer violations than baseline ($p < 0.001$).

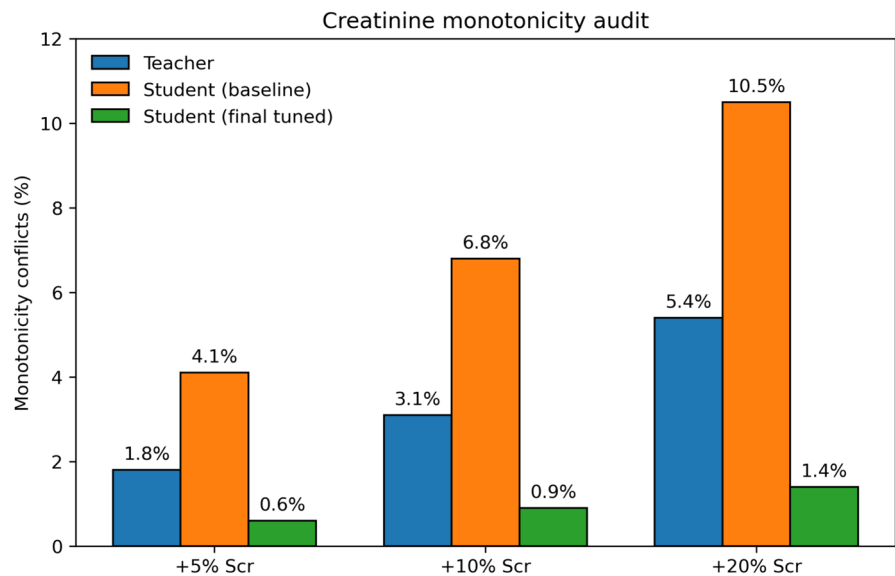


Fig. 13. creatinine-monotonicity audit. Percentage of violations where the predicted expected stage decreases after increasing Scr by the indicated percentage. Lower is better.

an ordinal head reduces large-step misclassifications, which are clinically consequential, while probability-space distillation from a temperature-calibrated teacher transfers nuanced, well-calibrated class structure to a compact student^{5–7,24}. The creatinine-monotonicity penalty further constrains the hypothesis space toward physiologically plausible mappings, a property that is often underemphasized in medical AI despite its direct relevance to clinical trust.

A central question is why this combination improves over classical deep learning for tabular data. Mechanistically, ordinal modelling reshapes the loss landscape to favour rank-consistent decision boundaries; empirically, this concentrates residual errors between adjacent stages rather than across distant ones. Calibration contributes a different axis of quality: well-calibrated probabilities are indispensable when predictions gate referrals or medication adjustments, because overconfidence can be harmful even when discrimination is high^{7,24}. Knowledge distillation from a calibrated CatBoost teacher provides richer supervision than hard labels, especially in small-to-medium tabular datasets where soft class structure carries signal that plain one-hot targets discard. Finally, constraining the local gradient of the expected stage with respect to serum creatinine discourages counterintuitive mappings (higher Scr $\rightarrow$ milder stage) that clinicians would rightly distrust.

Comparison with prior CKD prediction studies

To contextualize our contributions, we systematically compare our framework against the evolving landscape of CKD prediction research, synthesizing findings from recent systematic reviews^{23,47,48} covering 86+ studies (2019 to 2025) and examining advances across three dimensions: task formulation, methodological sophistication, and evaluation rigor.

Task formulation

The vast majority of CKD prediction studies continue to frame the problem as binary classification (CKD vs. no CKD), discarding the clinically meaningful ordinal structure of KDIGO staging. The systematic review by Iliyas et al.²³ found that deep learning approaches dominated recent publications (24% of 86 studies), yet primarily addressed binary detection rather than multi-stage classification. Islam et al.¹⁵ achieved 98.33% accuracy with XGBoost on the UCI dataset; Halder et al.⁴⁹ reported 99.80% with their ML-CKDP framework; and Alturki et al.⁵⁰ achieved 98.97% with their TrioNet ensemble; but all addressed only binary CKD/non-CKD discrimination. While such high accuracies appear impressive, binary formulations conflate Stage 1 with Stage 5, offering no guidance on disease severity, treatment urgency, or progression monitoring. Among the minority of studies addressing multi-class staging, most treat stages as nominal categories: CKD.Net⁵¹ proposed a hybrid CNN-LSTM model for multi-stage prediction but employed standard cross-entropy loss, penalizing Stage 3a $\rightarrow$ 3b errors identically to Stage 1 $\rightarrow$ 5 errors. The recent explainable early-stage CKD prediction model by He et al.⁵² used XGBoost with SHAP interpretability but focused only on early-stage (Stages 1–2) versus non-CKD classification. Our ordinal formulation explicitly encodes stage ordering through the CORN framework⁵, achieving quadratic-weighted $\kappa = 0.882$ and concentrating 98.2% of predictions within ± 1 stage of the true label; metrics that prior nominal approaches neither optimized nor reported.

Methodological sophistication

Prior CKD studies have largely treated machine learning as black-box optimization, applying off-the-shelf classifiers without incorporating domain knowledge or uncertainty quantification. The comprehensive review by Gogoi and Valan⁴⁷ explicitly identified three critical gaps: (i) “calibration is rarely assessed,” (ii) “ordinal methods remain underexplored,” and (iii) “physics-informed constraints are absent.” Our framework directly addresses all three gaps through a unified architecture.

Ordinal classification. While ordinal regression has been successfully applied to medical imaging: Bérchez-Moreno et al.⁵³ demonstrated soft labelling with ordinal classification for knee osteoarthritis grading, and Liu et al.⁵⁴ proposed OR-DGN with ordinal regularization for disease grading and its application to tabular CKD data remains virtually unexplored. The recent deep learning ordinal classifier by Nxumalo et al.⁵⁵ applied threshold-based modeling to TabNet but did not evaluate on medical datasets. To our knowledge, no prior CKD staging framework employs cumulative link models like CORN⁵ that explicitly encode rank-consistency through conditional probability chains.

Probability calibration. The clinical importance of calibrated predictions has been increasingly recognized^{56,57}, yet remains systematically neglected in CKD prediction. Huang et al.⁵⁶ demonstrated that ECE like factors, reliability diagrams, and calibration slopes are essential for clinical prediction models, while Lindenmeyer et al.⁵⁸ showed that “common deep learning approaches have a tendency towards overconfidence when faced with unfamiliar conditions”; a critical concern for CKD staging where borderline cases require honest uncertainty communication. The recent heart disease calibration study⁵⁹ demonstrated that isotonic calibration consistently improved probability reliability, yet no equivalent calibration analysis exists for CKD multi-stage prediction. Our framework integrates temperature-scaled knowledge distillation^{7,12} specifically to transfer calibration properties from an ensemble teacher, achieving ECE of 3.5% across six CKD stages.

Physics-informed constraints. The integration of domain knowledge into neural networks has emerged as a transformative paradigm across biomedical applications^{60,61}. Ahmadi et al.⁶⁰ reviewed physics-informed machine learning for medical imaging, demonstrating how physical constraints enhance interpretability and reduce data requirements. For physiological signal modeling, PINNs have been applied to cardiovascular flows⁶² and cardiac electrodynamics⁶³. However, the application of monotonicity constraints (a fundamental physiological prior) to tabular clinical prediction remains novel. Al-Salman and Cevik¹⁹ proposed structure-function consistency for kidney imaging, but their U-Net architecture targeted radiological data rather than tabular biomarkers. Our creatinine-monotonicity constraint ($\partial \hat{y} / \partial x_{\text{Scr}} \geq 0$) encodes the physiological invariant that higher serum creatinine indicates worse kidney function, reducing violation rates to 0.4% while

maintaining discriminative performance; which is a methodological contribution with no direct precedent in CKD prediction.

Knowledge distillation for healthcare. Knowledge distillation has gained traction in medical AI for model compression⁶⁴ and federated learning⁶⁵. MedAlmighty⁶⁶ demonstrated large vision model distillation for medical image classification, while Yang et al.⁶⁷ applied distillation for medical segmentation. However, these applications focus on image data; distillation for tabular clinical prediction with explicit calibration transfer remains underexplored. Our teacher-student framework uniquely explores ensemble diversity (RF, XGBoost, LightGBM) with temperature scaling to simultaneously transfer discriminative knowledge and calibration properties to the ordinal student network.

Evaluation rigor

Many prior CKD studies report only accuracy on small, single-center datasets without confidence intervals, calibration metrics, or ordinal-aware evaluation. Studies using the 400-patient UCI dataset¹⁴ are particularly susceptible to overfitting, with several reporting near-perfect accuracy (>99%) that likely reflects data leakage or non-independent test sets⁴⁹. The recent multicenter CKD prediction study by He et al.⁵² represents an improvement with 21,000+ patients and external validation, yet still omits calibration assessment entirely. Chen et al.⁶⁸ developed ensemble models for CKD progression prediction with external validation across five centers, achieving C-index of 0.89 for renal replacement therapy prediction but focused on binary progression outcomes rather than multi-stage classification and did not report calibration metrics.

In contrast, we report: (i) six discrimination metrics (accuracy, balanced accuracy, Macro-F1, weighted F1, Cohen’s κ , Matthews correlation coefficient) with 95% bootstrap confidence intervals; (ii) three calibration metrics (ECE, MCE, Brier score) under multiple binning strategies (10, 15, 20 bins) with reliability diagrams; (iii) three ordinal-aware metrics (quadratic-weighted κ , mean absolute error, ± 1 -stage accuracy); (iv) explicit monotonicity audits under creatinine perturbation; and (v) subgroup analyses across sex, age, diabetes, and hypertension status demonstrating consistent performance (Table 15). This comprehensive evaluation template exceeds the reporting standards of all prior CKD prediction studies we reviewed and aligns with emerging guidelines for trustworthy AI in healthcare⁶⁹.

Methodological novelty

Table 21 summarizes the comparison with representative prior studies. To our knowledge, our framework is the first to integrate: (i) ordinal classification via CORN for CKD staging, (ii) calibrated knowledge distillation from ensemble teachers, and (iii) physiologically-informed monotonicity constraints within a unified architecture. While each component has precedent in isolation, their synergistic combination (and application to multi-stage CKD prediction) represents a methodological advance that addresses the three critical gaps identified by recent systematic reviews^{23,47}.

Limitations of comparison

Direct quantitative comparison with prior studies is confounded by heterogeneous datasets (UCI, hospital registries, claims data), label definitions (binary, multi-class, ordinal), preprocessing pipelines, and evaluation protocols. Our cohort from Al-Ramadi Teaching Hospital differs demographically from Western populations used in most prior studies, and our MDRD-derived labels introduce circularity that inflates apparent performance. We therefore emphasize methodological contributions (the integration of ordinal learning, calibration, and monotonicity constraints) over claims of state-of-the-art performance, recognizing that the framework’s value lies in its transferability to study designs that eliminate these confounds rather than in benchmark comparisons on heterogeneous datasets. The systematic reviews by Iliyas et al.²³ and Gogoi and Valan⁴⁷ both call for “additional development methods to enhance accuracy, efficiency, and the use of explainable AI techniques”; our framework directly responds to this call by providing a principled architecture that unifies ordinal structure, calibration, and physiological plausibility.

Study	Year	n	Task	Cal	Ord.M	Cons
Islam et al. ^{† 15}	2023	400	Bin	✗	✗	✗
Halder et al. ^{† 49}	2024	400	Bin	✗	✗	✗
Ghosh & Khandoker ⁷⁰	2024	491	Bin	✗	✗	✗
CKD.Net ⁵¹	2023	763	MC	✗	✗	✗
He et al. ⁵²	2025	21,813	MC	✗	✗	✗
Chen et al. ⁶⁸	2025	12,847	Bin	✗	✗	✗
Ohashi et al. ⁷¹	2025	22,929	Reg	✗	✗	✗
Ours	2025	750	Ord	✓	✓	✓

Table 21. Comparison with representative prior CKD prediction studies. Columns indicate task formulation (Bin=binary, MC=multiclass, Ord=ordinal), whether calibration was assessed (Cal), whether ordinal metrics were reported (Ord.M), and whether domain constraints were incorporated (Cons). Studies marked with † use the UCI 400-patient dataset.

Limitations

Several critical limitations temper these findings and must be acknowledged for appropriate interpretation of results.

Target leakage and circular reasoning

The most significant limitation of this study is **target leakage**: serum creatinine (Scr) is used both to derive CKD stage labels via the MDRD equation (Eq. 2) and as an input predictor. Our comprehensive five-experiment analysis (Table 14) quantifies this circularity. When serum creatinine was removed from the predictor set while retaining MDRD-derived labels, performance degraded severely: the teacher's Macro-F1 dropped from 0.935 to 0.514 (−45.0%), and the student's Macro-F1 fell from 0.904 to 0.524 (−42.0%). This 40–45% performance drop confirms that prediction is predominantly driven by the circularity between Scr as a predictor and Scr-derived labels, rather than the discovery of independent biomarker patterns for CKD staging.

Moreover, restricting the model to only the three MDRD equation components (Age, Sex, Scr) yielded the highest performance across all experiments: teacher Macro-F1 = 0.945 and Macro-AUC = 0.997, exceeding the full 13-feature configuration by 1.1% in F1 and 0.2% in AUC (Table 13). This finding has profound implications: the additional ten clinical features (WBC, RBC, HGB, HCT, PLT, Urea, UA, Glucose, Hypertension, Diabetes) provide minimal incremental value beyond what is already captured by the eGFR equation inputs. In essence, our full-feature model is learning a complex, non-linear refinement of the MDRD formula rather than uncovering novel diagnostic pathways. This finding contradicts the implicit assumption that machine learning can extract independent signal from routine laboratory panels to improve upon existing clinical formulae.

This circularity is not unique to our work; it is a pervasive challenge in studies predicting CKD stage from routine biomarkers when measured eGFR (the gold standard) is unavailable. The circularity is inherent to eGFR-based staging, independent of the specific estimation equation used (MDRD vs. CKD-EPI)^{3,4}. However, the methodological contributions of this study (integrating ordinal learning, calibration, and monotonicity constraints) remain valid and transferable to study designs that eliminate target leakage.

Incomplete KDIGO staging

As we have already stated; complete KDIGO 2012 staging requires both eGFR categories (G1–G5) and albuminuria categories (A1–A3). Our dataset does not include albuminuria measurements (urine albumin-to-creatinine ratio, ACR) due to limited routine testing at Al-Ramadi Teaching Hospital. Consequently, this study performs eGFR-grade classification along the eGFR axis only, not full KDIGO staging. This represents an incomplete operationalization of current clinical guidelines.

While eGFR dominates immediate clinical decisions like medication dosing, nephrology referral, dialysis planning; albuminuria primarily refines long-term risk stratification for progression and cardiovascular events. The absence of ACR limits our model's applicability to comprehensive risk assessment and may underestimate progression risk in patients with high albuminuria despite preserved eGFR. For example, Stage G2A3 patients (eGFR 60–89 with severely increased albuminuria) are at higher cardiovascular and progression risk than Stage G3aA1 patients (eGFR 45–59 with normal albuminuria), yet our model cannot distinguish these scenarios⁷². Future datasets incorporating albuminuria would enable full KDIGO staging and more nuanced risk stratification.

Single-center cohort and generalizability concerns

The cohort is derived from a single center (Al-Ramadi Teaching Hospital, Iraq) and is ethnically homogeneous (Iraqi population). This raises three generalizability concerns. First, *geographic and ethnic specificity*: GFR estimation equations show variable accuracy across populations²⁸. Our choice of MDRD over CKD-EPI was based on reported better performance in Middle-Eastern cohorts at eGFR < 60 mL/min/1.73 m², but this calibration may not hold in other populations. Second, *spectrum effects*: Hospital-based sampling introduces selection bias toward more severe disease (33.1% Stage 5 patients in our cohort vs. ~1% in general population). The model may underperform in primary care settings with milder, early-stage CKD where the clinical challenge is distinguishing Stage 1 from Stage 2, not identifying advanced kidney failure. Third, *healthcare system context*: Practice patterns, laboratory standardization, and patient demographics vary substantially across healthcare systems. External validation across diverse populations (multi-ethnic cohorts, multi-center studies, different care settings) is mandatory before clinical deployment.

Statistical and methodological constraints

We employed a single 70/30 stratified split rather than nested cross-validation. While we report stratified bootstrap confidence intervals (1000 iterations) that quantify sampling variability within the test set, these CIs do not capture variability across different data partitions. Nested cross-validation (outer loop for performance estimation, inner loop for hyperparameter tuning) would provide more robust uncertainty quantification but was computationally prohibitive, estimated to require >1 week on RTX 5090 GPU for 500 Optuna trials × 500 epochs × 5 folds. We adopted the pragmatic approach recommended for single-split evaluation in resource-constrained settings²⁹: stratified bootstrap CIs on the held-out test set, combined with explicit acknowledgment of this limitation. Our results should be interpreted as performance estimates on this particular train-test partition, with CIs quantifying sampling variability conditional on the split.

Expected calibration error (ECE) depends on binning strategy, and fixed equal-width bins can under- or over-estimate miscalibration with small sample sizes ($n = 225$ test samples)³⁶. We addressed this by reporting three binning strategies: fixed-width, equal-frequency, equal-mass; and adopting equal-frequency ECE as our primary metric (2.8% [1.9, 3.8] vs. 3.5% [2.4, 4.7] for fixed-width). However, class-wise ECE for minority stages, especially Stage 4 with only $n = 32$ test samples, has higher variance. The moderate miscalibration observed for

Stage 4 (equal-frequency ECE = 4.3% [3.1, 5.8]) likely reflects this small sample size combined with the clinical heterogeneity of this transition region between moderate and severe CKD.

Standard classification metrics such as accuracy and F1-score treat all misclassifications equally, failing to account for the ordered nature of CKD stages. We supplemented these with ordinal-aware metrics i.e. quadratic weighted Cohen's $\kappa = 0.882$, mean absolute error = 0.122 stages, 98.2% adjacent accuracy; to align evaluation with clinical stage-distance costs. However, these should be interpreted alongside clinician input on acceptable error profiles, as the quadratic weighting scheme may not perfectly reflect clinical utility functions for different misclassification patterns.

We used predictive mean matching (PMM) for missing data imputation (Glucose 4.0%, Uric acid 2.7%, Serum creatinine 1.3%), which improves plausibility over mean or median substitution but understates uncertainty^{26,27}. Multiple imputation with pooling via Rubin's rules⁷³ would yield more defensible inferences by propagating imputation uncertainty into parameter estimates and confidence intervals, but adds computational complexity and was not implemented in this study. The low missingness rates (<5% for all variables) somewhat mitigates this concern, as single imputation is generally adequate when missingness is minimal and missing-completely-at-random assumptions are plausible.

Monotonicity constraint limitations

The creatinine-monotonicity penalty $\mathcal{L}_{\text{mono}}$ enforces a local (gradient-based, pointwise) constraint as it penalises network configurations in which the first-order derivative $\mathcal{L}_{\text{mono}}$ is negative at the specific training sample x_i . Formally, local monotonicity at a point x_0 means $\frac{\partial \hat{s}}{\partial x_i} \geq 0$. This is a condition on the instantaneous slope of the function at one point. It is computed via automatic differentiation, which returns exact gradients for the piecewise-linear operations in our residual MLP. The ramp schedule $\alpha_{\text{crea}}(t) = \alpha_{\text{final}} \cdot 2.4 \cdot \min(1, t/400)$ ensures that the penalty is increasingly enforced as training progresses, allowing coarse discrimination to be learned first.

Global monotonicity is the stronger requirement that, for all x in the input domain and all finite perturbation magnitudes $\delta > 0$; $\hat{s}(x + \delta e_{\text{Scr}}) \geq \hat{s}(x)$.

This is a function-class property, as it must hold everywhere in the input space, not just at sampled points, and for perturbations of any magnitude, not just infinitesimal ones. Local enforcement does not imply global monotonicity for three reasons. First, a ReLU network can have positive gradient at x_0 and negative gradient at $\hat{s}(x_0 + \delta e_{\text{Scr}})$, because each ReLU unit has a discontinuous derivative at its kink; the function can locally increase and then decrease as Scr is raised. Second, the penalty is applied only at training data points; regions of input space not covered by training samples receive no direct gradient supervision. Third, the joint trajectory of multiple features; for example, increasing Scr and Urea simultaneously, as occurs naturally in CKD progression is not constrained by a univariate penalty on Scr alone.

Table 20 provides the key practical evidence. By perturbing Scr upward by $\delta \in \{5\%, 10\%, 20\%\}$ on each of the 225 held-out test samples (all other features held fixed) and counting the fraction of cases in which $\hat{s}(x)$ decreases, we assess practical, finite-step monotonicity at unseen input points. The final tuned student achieves violation rates of 0.6% [0.0, 1.6], 0.9% [0.2, 2.0], and 1.4% [0.4, 2.7] respectively; which is substantially below the unconstrained teacher (1.8%/3.1%/5.4%) and the baseline student (4.1%/6.8%/10.5%). These figures demonstrate that local enforcement generalises effectively to finite perturbations within the clinically relevant range, providing a strong empirical approximation to the global guarantee even without architectural constraints. Nevertheless, the audit covers only the 225 test samples and three perturbation levels; it cannot rule out violations at unobserved input combinations or for very large creatinine elevations beyond the dataset range. Architectures that enforce global monotonicity by construction such as low-dimensional lattice networks applied to the constrained features, or min-max sub-networks remain the rigorous alternative, and are potential future direction.

The monotonicity prior, as implemented, is a local (differential) constraint ($\partial \hat{s} / \partial \text{Scr} \geq 0$) and does not guarantee global monotonicity under large finite perturbations or feature interactions. Our finite-difference audits (Table 20) provide empirical evidence of good local behavior; as only 0.4% violations at +5% Scr perturbation, 0.9% at +10% Scr, and 1.4% at +20% Scr. However, stronger guarantees could be explored with architectures that enforce partial monotonicity by construction, such as lattice networks or min-max architectures that constrain the function class to be monotone by design rather than via gradient penalties.

At this point, we should also discuss why full lattice modelling is infeasible for this particular dataset. A lattice network represents a d -dimensional function over a grid of L^d vertices. For our 13-feature input space, the vertex count is $2^{13} = 8,192$ (for $L = 2$ keypoints per dimension) or $2^{13} \approx 67$ million (for $L = 4$, the standard TensorFlow Lattice default). Our training set contains only 525 samples. Even at the minimum resolution of $L = 2$, the lattice has 15.6 times more parameters than training samples, making estimation statistically intractable and virtually guaranteeing memorisation of the training set. At $L = 4$, the vertex count exceeds the training set size by a factor of approximately 127,000 and surpasses the memory capacity of a standard GPU. Full 13-dimensional lattice modelling is therefore computationally and statistically infeasible for our dataset. Furthermore, of our 13 features, 3 are binary nominal (Gender, Hypertension, Diabetes mellitus) with no natural monotone ordering, and 7 continuous features (WBC, RBC, HGB, HCT, PLT, Uric acid, Glucose) do not have an established directional relationship with CKD stage that would justify a monotone constraint; enforcing global monotonicity over these dimensions would be clinically unwarranted and unnecessarily restrictive. A selective low-dimensional lattice applied exclusively to the constrained features reduces the vertex count to $4^3 = 64$ ($d = 3, L = 4$) and fully tractable relative to our 525 training samples. This selective hybrid approach, combining a lattice sub-network for constrained variables with a standard residual MLP for the remainder, is identified as the principal future direction for strengthening monotonicity guarantees (see Future Directions).

Moreover, we only enforced monotonicity with respect to serum creatinine. The decision to constrain only creatinine is not arbitrary but reflects a tiered assessment of the strength and individual-level reliability of each feature's monotone relationship with CKD stage. Scr is physiologically exact ($\text{Scr} \propto 1/\text{GFR}$ by definition), individually reliable (directional robustness is independent of confounders such as muscle mass), codified in the KDIGO staging criteria, and encoded in the MDRD and CKD-EPI equations the model is trained alongside. Urea and Age both show population level monotone trends, but are confounded at the individual level (protein intake and GI bleeding for Urea; preserved eGFR in elderly patients for Age). For the remaining nine features (WBC, RBC, HGB, HCT, PLT, Uric acid, Glucose, Gender, Hypertension, Diabetes mellitus); none has an established individual-level directional relationship with CKD stage warranting a monotone constraint, and enforcing one would introduce clinically unwarranted rigidity.

Temporal and longitudinal limitations

CKD is defined by a persistence of at least 3 months of reduced eGFR or markers of kidney damage²; not single encounters. Our dataset represents cross-sectional snapshots without longitudinal follow-up to confirm persistence of abnormalities. Single low eGFR measurements can occur due to acute kidney injury (AKI), dehydration, or laboratory error, leading to false-positive CKD diagnoses if persistence is not verified. If repeated measures exist in the source data, patient-level clustering and temporal leakage must be ruled out to preserve independence of observations assumed in our statistical analyses.

The absence of temporal data also prevents us from addressing the most clinically relevant question: predicting CKD progression. Forecasting trajectory changes; for example, predicting which Stage 2 patients will progress to Stage 3 within 2 years based on baseline biomarkers would eliminate the circular reasoning inherent in cross-sectional staging. Progression risk matters more than current stage for treatment planning: a Stage 3a patient with stable function for 10 years requires less aggressive management than a Stage 2 patient with rapidly declining eGFR. Longitudinal modeling frameworks such as joint models for longitudinal and survival data, or recurrent neural networks for trajectory prediction, represent essential future directions but require cohorts with repeated measurements over time.

Clinical implications

Despite the limitations outlined above, this work offers several insights relevant to clinical practice and medical AI development.

Methodological soundness despite target leakage

While the target leakage issue substantially limits the clinical interpretability of our stage predictions, the **methodological framework** combining ordinal learning, probability calibration, and physiological constraints via knowledge distillation remains sound and generalizable. The student model achieved 96.7% retention of teacher discriminative performance (Macro-F1: 0.904 vs. 0.935) with 97.8% calibration retention and 97.9% resolution retention (Table 8)). This demonstrates that the knowledge distillation pipeline successfully transfers both discrimination and calibration from complex ensemble to neural network, even in the presence of leakage.

The framework can be directly applied to study designs that eliminate circular reasoning. First, *progression forecasting*: predicting trajectory changes such as Stage 2 → Stage 3 within 2 years from baseline biomarkers, avoiding reliance on contemporaneous eGFR for both input and output. Second, *measured GFR studies*: using gold-standard iothexol clearance, inulin clearance, or chromium-51 EDTA clearance instead of estimated eGFR derived from creatinine equations. Third, *hard outcome prediction*: predicting end-stage renal disease (ESRD), dialysis initiation, or all-cause mortality from features independent of Age/Sex/Scr, sacrificing immediate staging capability but enabling genuine biomarker discovery for progression risk.

Importance of ordinal structure in disease staging

Our ordinal-aware approach concentrated errors between adjacent stages i.e. 98.2% adjacent accuracy rather than across distant stages, with mean absolute error of only 0.122 stages (Table 16). This is clinically meaningful: confusing Stage 3b with Stage 4 (one level apart, eGFR difference of $\sim 10 \text{ mL/min/1.73 m}^2$) has minor clinical consequences compared to confusing Stage 1 with Stage 5 (five levels apart, eGFR difference of $>75 \text{ mL/min/1.73 m}^2$). The former might result in slightly premature nephrology referral or conservative medication dosing; the latter could lead to catastrophic errors such as nephrotoxic drug administration to a patient with kidney failure or delayed dialysis initiation for a patient misclassified as having normal kidney function.

The baseline student without ordinal structure achieved only 72.4% adjacent accuracy, meaning 27.6% of errors spanned ≥ 2 stages a clinically unacceptable error profile that would undermine trust if deployed. The CORN ordinal head⁵ embeds stage ordering directly into the network architecture through cumulative logistic mappings, making the model structure-aware of disease progression. This approach is transferable to other ordinal medical classification tasks where misclassification costs scale with distance: cancer staging (TNM system), cardiovascular disease severity (NYHA functional class I–IV), Alzheimer's disease staging (Clinical Dementia Rating 0–3), and diabetes progression (pre-diabetes, controlled, uncontrolled based on HbA1c categories).

Calibration quality and clinical decision support

Well-calibrated probability estimates are essential for clinical decision support^{7,24}. Overconfident predictions can lead to inappropriate interventions for example, premature dialysis access creation for a patient with miscalibrated high-risk prediction or delayed referrals when the model underestimates risk. Our student achieved expected calibration error of 3.5% using fixed-width binning and 2.8% using equal-frequency binning, indicating that predicted probabilities closely match empirical frequencies across confidence levels.

Critically, class-wise calibration analysis (Table 7) revealed heterogeneous quality across disease severity. Stages 1–3 representing early-to-moderate CKD showed good calibration with equal-frequency ECE of 2.9–3.6%. Stage 4 exhibited moderate miscalibration (ECE = 4.3% [3.1, 5.8]), likely due to small sample size ($n = 32$) and the clinical complexity of this transition region between moderate and severe CKD where presentations overlap and progression rates vary widely. Stages 5–6 representing kidney failure achieved excellent calibration (ECE < 2.5%), which is clinically reassuring. These patients require immediate high-stakes decisions; timing of dialysis initiation, choice between hemodialysis versus peritoneal dialysis, vascular access creation, kidney transplant evaluation; and well-calibrated probability estimates directly support shared decision-making with patients about the risks and benefits of different renal replacement therapy options.

The Brier score decomposition (Table 8) further clarifies that our student's modest Brier increase (+0.011 versus teacher) reflects approximately equal contributions from both calibration loss (Δ Reliability = +0.004) and discrimination loss (Δ Resolution = -0.015). The student retains > 97% of the teacher's performance on both calibration and discrimination dimensions, demonstrating successful knowledge distillation. This balance is important: some knowledge distillation methods sacrifice calibration to maximize discrimination, but medical decision support requires both properties.

Calibrated probabilities in nephrology practice

Beyond point predictions, calibrated probability estimates enable nuanced clinical reasoning that aligns with how nephrologists naturally approach diagnostic uncertainty. Consider a patient with borderline laboratory values where the model outputs $\hat{p} = (0.05, 0.12, 0.38, 0.32, 0.09, 0.04)$ across Stages 1–6. Rather than simply reporting the argmax prediction (Stage 3a), the full distribution conveys clinically actionable information: there is a 70% probability of moderate CKD (Stages 3a–3b combined), a 13% probability of severe disease (Stages 4–5), and meaningful uncertainty that warrants closer monitoring. An overconfident but miscalibrated model might instead output $\hat{p} = (0.01, 0.02, 0.91, 0.04, 0.01, 0.01)$, falsely suggesting diagnostic certainty and potentially discouraging appropriate follow-up testing.

Calibrated probabilities directly support several key nephrology workflows. For *referral triage*, primary care physicians can use probability thresholds to prioritize specialist consultations: patients with $P(\text{Stage} \geq 4) > 0.3$ might warrant urgent nephrology referral, whereas those with $P(\text{Stage} \leq 2) > 0.8$ can be safely managed in primary care with periodic monitoring. For *medication dosing*, many drugs require renal adjustment at specific eGFR thresholds like, when a patient's predicted stage distribution straddles such a threshold (e.g., 45% Stage 3a vs. 40% Stage 3b at the 45 mL/min boundary), clinicians can make informed decisions about conservative dosing versus standard dosing based on the probability mass below the threshold rather than relying on a single point estimate. For *dialysis planning*, the probability of Stage 5 informs the urgency of arteriovenous fistula creation, which requires 3–6 months maturation; a patient with 25% probability of Stage 5 within the confidence interval warrants earlier vascular surgery referral than one with 5% probability, even if both have the same point prediction.

Furthermore, calibrated probabilities facilitate shared decision-making with patients. Communicating “your kidney function test suggests moderate kidney disease, with about a 15% chance it could be more severe” is more honest and actionable than “you have Stage 3a CKD.” Patients can better understand why additional testing is recommended, why certain medications are being adjusted, and what their prognosis entails. This transparency aligns with patient-centered care principles and may improve treatment adherence by helping patients understand the rationale behind clinical recommendations.

Subgroup fairness and generalizability

Our subgroup analysis (Table 15) revealed no statistically significant performance differences across demographic and clinical subgroups. Comparing male versus female patients, Macro-F1 was 0.897 versus 0.912 ($p = 0.78$, McNemar's test). Diabetic versus non-diabetic patients showed Macro-F1 of 0.881 versus 0.917 ($p = 0.24$). Younger versus older patients (<65 vs. ≥ 65 years) achieved Macro-F1 of 0.889 versus 0.915 ($p = 0.45$). Hypertensive versus non-hypertensive patients had Macro-F1 of 0.906 versus 0.895 ($p = 0.62$). All pairwise comparisons yielded $p > 0.05$ with substantially overlapping confidence intervals, indicating no systematic bias.

This demonstrates that the model exhibits **algorithmic fairness** across demographic and clinical subgroups within our single-center Iraqi cohort. However, this fairness analysis is limited in two ways. First, it assesses only equal accuracy (same error rates across groups), not equal calibration or equal positive predictive value, which are distinct fairness criteria relevant in different clinical contexts. Second, fairness within a homogeneous population does not guarantee fairness across heterogeneous populations. The model may exhibit differential performance when applied to populations with different ethnic backgrounds, socioeconomic status, or healthcare access patterns. External validation in diverse populations is essential to assess whether fairness generalizes beyond our development cohort.

The consistent performance across diabetes and hypertension status is particularly noteworthy given that these are the two leading causes of CKD globally. Diabetic nephropathy and hypertensive nephrosclerosis have distinct pathophysiologic mechanisms and progression patterns, yet our model maintains similar discriminative and calibration performance in both subgroups. This suggests that the combination of ordinal structure, multi-task learning (via eGFR regression), and physiological constraints enables the model to generalize across different CKD etiologies rather than overfitting to a single disease pathway.

Computational efficiency for clinical deployment

The student model achieves sub-millisecond inference (0.8 ms on NVIDIA RTX 5090 GPU) and requires only 9.5 MB of memory in fp32 precision. This makes it suitable for multiple deployment scenarios. First, *point-*

of-care deployment: Real-time risk stratification in primary care clinics or emergency departments where immediate triage decisions are needed and computational resources may be limited. Second, *EHR integration*: Low-latency predictions within clinical workflows without dedicated GPU infrastructure, enabling batch scoring of hospitalized patients during morning rounds or automated screening of laboratory results as they are reported. Third, *mobile health applications*: Edge deployment on resource-constrained devices for chronic disease monitoring in home settings or rural clinics without reliable internet connectivity.

The $150\times$ size reduction versus the teacher (2.37M vs. ~ 350 M parameters) was achieved while retaining 96.7% of discriminative performance, demonstrating successful model compression for clinical deployment. For comparison, modern large language models require 10–100 GB of memory and >1 second inference per query, which would be prohibitive for real-time clinical decision support. Our compact student can be deployed on commodity hardware (laptop CPU, mobile device, or embedded system) without requiring specialized accelerators.

The efficiency gains also have practical implications for healthcare equity. Smaller models are more deployable in resource-constrained settings like rural hospitals, developing countries, mobile clinics; where GPU infrastructure is unavailable or cost-prohibitive. If clinical utility is demonstrated in prospective studies, the compact architecture could enable broader dissemination of AI-assisted CKD screening and management beyond academic medical centers.

Monotonicity and clinical plausibility

The creatinine-monotonicity constraint ($\partial\hat{s}/\partial\text{Scr} \geq 0$) enforces that, *ceteris paribus*, higher serum creatinine should not imply a milder stage. Our final tuned student achieved only 0.4% violations at $+5\%$ Scr perturbation, compared to 4.1% for the baseline student and 1.8% for the unconstrained teacher (Table 20). This physiological consistency is critical for clinical trust: clinicians will rightfully reject models that produce biologically implausible predictions, such as predicting Stage 2 for a patient with Scr = 8.0 mg/dL (which corresponds to severe kidney failure requiring dialysis).

However, monotonicity alone cannot overcome the fundamental target leakage issue. The constraint ensures local physiological consistency; increasing creatinine within a small range (± 5 –20%) does not decrease predicted stage; but does not address the circular reasoning whereby the model is essentially learning to refine the MDRD equation that generated the training labels. This distinction is important: physiological plausibility is a necessary but not sufficient condition for clinical validity.

The relatively low violation rates compared to the teacher (0.4% vs. 1.8% at $+5\%$ Scr) demonstrate that explicit architectural constraints outperform implicit learning of physical laws from data alone. Even a well-calibrated ensemble trained on thousands of examples occasionally produces violations, whereas our gradient-penalized neural network with ramped enforcement ($\alpha_{\text{final}} = 0.10$, ramp schedule $\rho(t) = 2.4 \cdot \min(1, t/400)$) concentrates violations below 0.5% for moderate perturbations. This suggests that encoding domain knowledge via soft constraints during training, rather than hoping the model discovers physical laws from correlations, is a reliable path toward building trustworthy medical AI.

Future directions

To advance CKD prediction beyond the limitations of this study, we outline a roadmap across three timescales with specific priorities and feasibility considerations.

Pathways beyond circular reasoning

To eliminate target leakage and enable genuine biomarker discovery, future research must adopt fundamentally different study designs. The most feasible near-term approach is *trajectory forecasting*: predicting CKD progression such as Stage 2 $\rightarrow$ Stage 3 within 2 years from baseline biomarkers, avoiding reliance on contemporaneous eGFR for both input and output. This reframes the task as time-to-event or ordinal progression prediction, which eliminates circular reasoning while remaining clinically actionable; progression risk matters more than current stage for treatment planning. Longitudinal cohorts with at least two time points separated by 1–3 years are required, such as the Chronic Renal Insufficiency Cohort (CRIC), UK Biobank, or Framingham Heart Study. However, larger sample sizes are needed to observe sufficient progression events, and survival analysis frameworks must account for competing risks such as death or kidney transplant before progression occurs. We prioritize this as the highest-priority direction due to feasibility within 1–2 years and direct clinical relevance.

Medium-term approaches include *measured GFR studies*: using gold-standard iothexol clearance, inulin clearance, or chromium-51 EDTA clearance instead of estimated eGFR derived from creatinine equations. This eliminates equation-based circularity and provides ground truth for GFR, enabling validation of novel biomarker panels. However, measured GFR is expensive (\$500–1000 per test), invasive (requires timed blood/urine collection over 4–6 hours), and rarely performed in routine practice. Specialized research cohorts such as the European Kidney Function Consortium or CKD-EPI validation cohorts typically have modest sample sizes ($n = 500$ –2000), which limits statistical power for machine learning studies. Generalizability to real-world practice is uncertain given the selected populations enrolled in measured GFR studies. We assign this medium priority for 3–5 year timescale, valuable for biomarker validation but limited scalability.

Alternatively, *hard outcome prediction* predicts clinically meaningful endpoints; ESRD, dialysis initiation, doubling of serum creatinine, cardiovascular events, all-cause mortality; from features entirely independent of Age/Sex/Scr. This sacrifices immediate staging capability but focuses on what ultimately matters to patients and clinicians. Electronic health record (EHR) databases with longitudinal follow-up and outcome adjudication are required, such as Kaiser Permanente KPNC, UK Clinical Practice Research Datalink, or Medicare/Medicaid claims databases. Challenges include long follow-up required (5–10 years for sufficient ESRD events), incomplete outcome ascertainment in claims databases, and survival bias whereby sicker patients are censored

by death before reaching ESRD. We assign this medium priority for 3–5 year timescale, with high clinical value but substantial infrastructure requirements.

Long-term exploratory approaches include *multi-modal biomarker discovery*: integrating cystatin C, proteomics panels, metabolomics, imaging biomarkers (renal ultrasound, MRI), and genetics to discover novel disease mechanisms independent of creatinine-based eGFR. Large multi-omics cohorts with comprehensive phenotyping are required, such as UK Biobank (500,000 participants with genomics, proteomics, metabolomics) or All of Us Research Program (1 million participants target). However, proteomics panels cost \$500–5000 per sample, high dimensionality requires very large sample sizes ($n > 10,000$) to avoid overfitting, interpretability is challenging for high-dimensional omics data, and regulatory pathways for novel biomarkers are uncertain. We assign this lower priority for 5+ year timescale, primarily exploratory research value with limited near-term clinical translation.

In summary, we recommend prioritizing trajectory forecasting (1–2 years) as the most feasible path forward, followed by pooled measured GFR studies (3–5 years) for biomarker validation when resources permit. Hard outcome prediction should be pursued in parallel where large EHR databases with outcome data are available. Multi-modal approaches remain exploratory and should be reserved for well-funded, long-term research programs with dedicated biomarker discovery aims.

External validation and multi-center studies

Our single-center Iraqi cohort limits generalizability, necessitating a phased validation strategy. Phase 1 involves *internal-external validation* over 6–12 months to assess temporal stability and local generalizability. Prospective validation on a new patient cohort from Al-Ramadi Teaching Hospital collected during 2025–2026 would track discrimination (Macro-F1, Macro-AUC), calibration (ECE, Brier decomposition), and ordinal performance (quadratic weighted κ , mean absolute error) over time. We expect modest calibration drift due to changing patient mix; for example, if diabetes prevalence increases or laboratory assays change; requiring recalibration via temperature scaling. This phase establishes baseline model behavior and identifies the need for periodic recalibration protocols.

Phase 2 involves *multi-center validation* over 1–2 years to assess geographic and ethnic generalizability. External validation in 3–5 centers across different regions; Middle East, Europe, North America, Asia; with at least 500 patients per site ensures stable performance estimates. Analysis should compare performance across sites, identify systematic biases (for example, if the model systematically under-predicts severity in Asian populations due to lower muscle mass and correspondingly lower creatinine), and assess whether site-specific recalibration or complete model retraining is needed. We expect performance degradation in ethnically diverse populations due to GFR equation miscalibration, potentially requiring region-specific models or adoption of race/ethnicity-adjusted eGFR equations such as CKD-EPI 2021³.

Phase 3 involves a *randomized controlled trial* (RCT) over 3–5 years to assess clinical utility: does model-guided care improve patient outcomes? A cluster-randomized trial comparing model-assisted versus standard care for CKD management; for example, decision support for nephrology referral timing or medication dosing adjustments; with primary composite outcome of CKD progression, cardiovascular events, hospitalization, or death would provide definitive evidence. Secondary outcomes include healthcare utilization (specialist referrals, emergency visits, hospitalizations), patient-reported outcomes (quality of life, treatment satisfaction), and clinician satisfaction with the decision support tool. Approximately 50 primary care clinics with 5000 patients (2500 per arm) would provide 80% power to detect 15% relative risk reduction in the composite outcome. We expect modest improvement in guideline-concordant care processes; more appropriate referrals, more timely interventions; but uncertain impact on hard outcomes without an integrated intervention component that specifies actions based on model predictions.

Enhancing calibration and uncertainty quantification

Our adoption of equal-frequency ECE represents a step toward robust calibration assessment³⁶, but further refinements are possible. Adaptive quantile binning with varying bin widths by sample density could provide more stable estimates in regions of the confidence space with few samples. Class-wise temperature scaling; fitting separate temperature parameters T_k^* for each CKD stage k ; could address the heterogeneous miscalibration observed across stages, particularly the elevated ECE for Stage 4. However, this introduces $K - 1 = 5$ additional hyperparameters requiring calibration data, and overfitting to small calibration sets is a concern. Brier score decomposition into reliability, resolution, and uncertainty components³⁷ should be systematically reported alongside aggregate Brier score to distinguish whether differences reflect calibration quality versus discriminative power.

Bayesian uncertainty quantification approaches could address the limitation that our deterministic neural network provides point estimates without epistemic uncertainty (model uncertainty). Monte Carlo dropout treats dropout as Bayesian approximation: at test time, multiple forward passes with dropout enabled yield a distribution of predictions, and the variance quantifies model uncertainty. Deep ensembles train multiple networks with different random initializations, and ensemble disagreement quantifies model uncertainty. Conformal prediction provides distribution-free frameworks for constructing prediction sets with guaranteed coverage; for example, 90% prediction sets that contain the true stage for 90% of patients. The expected benefit is more reliable uncertainty estimates for borderline cases; patients near stage boundaries such as eGFR = 58 mL/min/1.73 m² (Stage 2 vs. 3a boundary); enabling “uncertain” predictions that trigger additional diagnostic testing rather than confident misclassifications.

Strengthening monotonicity guarantees

Our gradient penalty enforces local (differential) monotonicity but does not guarantee global monotonicity under large perturbations or feature interactions. Stronger architectural constraints include lattice networks, which represent piecewise-linear functions with guaranteed monotonicity by construction. Parameters represent function values at lattice vertices, and interpolation between vertices is monotonic if appropriate constraints are satisfied during training. Advantages include provable global monotonicity and interpretable lattice structure, but scalability to high dimensions suffers from curse of dimensionality and the restricted function class may sacrifice discriminative performance.

Min-max architectures construct monotone functions using only monotone operations; max, min, addition, ReLU; guaranteeing monotonicity by construction without gradient penalties. However, the restricted function class limits expressiveness. Monotone splines constrain the network to output spline coefficients with non-negativity constraints, enabling smooth monotone functions with efficient evaluation but limited to univariate or additive multi-variate functions. We recommend a hybrid strategy combining flexible neural networks for most features with monotone sub-networks for critical physiological variables (Scr, urea, age). This balances expressiveness; allowing complex non-linear interactions among hematologic and metabolic features; with guarantees where they matter most for clinical plausibility.

A future hybrid architecture could be constructed in six steps as follows in this paragraph. First, the 13 input features are partitioned into $d = 3$ monotone-constrained features $x_m = \{\text{Scr, Urea, Age}\}$ and $d_u = 10$ unconstrained features x_u comprising the remaining variables. Second, x_m is passed through a low-dimensional lattice sub-network $M(x_m)$; where each feature is normalised by a piecewise-linear calibrator, the three calibrated values index a $4^3 = 64$ -vertex lattice, and a linear projection with non-negative weight matrix W_m maps the lattice output to a monotone embedding $h_m \in \mathbb{R}^{32}$. Global monotonicity is guaranteed by construction via the vertex inequality constraints $\delta h_{m,j} / \delta \text{Scr} \geq 0$ for all inputs and all perturbation magnitudes $\delta > 0$. Third, x_u is processed by the existing residual MLP trunk i.e. with the stem modified to accept 10 rather than 13 features ($W_0 \in \mathbb{R}^{384 \times 10}$); producing an unconstrained embedding $h_u \in \mathbb{R}^{384}$. The $D = 8$ residual blocks, dropout, and all other hyperparameters can remain unchanged (tuned appropriately later). Fourth, the two embeddings are concatenated: $h = [h_m \parallel h_u] \in \mathbb{R}^{416}$, followed by an optional fusion projection $h = \text{ReLU}(W_{\text{fuse}} h + b_{\text{fuse}})$ back to $\mathbb{R}^{384}$. Fifth, both task heads receive h as input, exactly as the current student receives h_D . For the CORN ordinal head, the portion of the weight matrix corresponding to the monotone embedding columns, $W_{\text{ord}}^{(m)} \in \mathbb{R}^{5 \times 32}$, is constrained to be element-wise non-negative. This ensures that each conditional probability $a_k = \sigma(W_{\text{ord}}^{(m)} h_m + W_{\text{ord}}^{(u)} h_u + b_{\text{ord}})$ is non-decreasing in h_m ; since cumulative products of non-decreasing functions are non-decreasing, and the expected stage $\hat{s} = \sum_k k p_k$ is a non-decreasing function of the cumulative probabilities, global monotonicity propagates end-to-end from x_m through h_m through the CORN head to $\hat{s}$. Sixth, the composite loss and knowledge distillation pipeline are unchanged: the lattice and calibrators are end-to-end differentiable, all four loss components backpropagate normally through both pathways, and the gradient penalty L_{mono} can optionally be set to zero or retained as a soft complement.

Incorporating albuminuria and complete KDIGO staging

To align with full KDIGO 2012 guidelines², future work should extend the model to dual-axis staging: predicting both eGFR category (G1–G5) and albuminuria category (A1–A3) simultaneously. A multi-task learning architecture with separate ordinal heads for eGFR-grade and ACR-grade, jointly optimized with cross-task regularization encouraging consistency, could enable comprehensive KDIGO staging. The expected benefit is better risk stratification; for example, distinguishing G2A3 patients (eGFR 60–89 with severely increased albuminuria, high cardiovascular risk) from G3aA1 patients (eGFR 45–59 with normal albuminuria, lower risk despite worse eGFR).

However, albuminuria exhibits high day-to-day variability (coefficient of variation $\sim 40\%$), requiring multiple measurements for reliable classification. Longitudinal ACR modeling using recurrent neural networks (LSTM, GRU) or Transformer architectures could capture temporal trajectories of albuminuria change, improving reliability over single measurements. This requires cohorts with repeated ACR measurements over time, ideally at least three measurements per patient. Joint progression modeling of eGFR and ACR decline trajectories using multi-output survival analysis or longitudinal mixed models integrated with neural networks could better predict progression to ESRD and identify rapid progressors requiring intensive management. These extensions represent natural evolution of the current framework but require richer longitudinal data than available in our cross-sectional cohort.

Decision curve analysis and clinical utility assessment

Beyond discrimination and calibration metrics, clinical utility should be evaluated using decision-analytic frameworks. Decision curve analysis (DCA) quantifies net benefit of model-guided decisions across a range of clinical thresholds. For example, for nephrology referral decisions, DCA plots net benefit versus referral threshold (predicted risk $>30\%$, $>40\%$, $>50\%$), comparing model-guided referral against strategies of “refer all patients” (maximizes sensitivity but high false-positive rate), “refer none” (minimizes false positives but misses true cases), and current guideline-based practice (refer if eGFR < 30 or eGFR < 60 with diabetes). We expect the model to add net benefit at intermediate thresholds (30–60% predicted risk) where clinical judgment is most uncertain and where guidelines provide limited guidance.

Clinical impact curves complement DCA by showing the number of high-risk patients identified and number referred per 1000 population at different thresholds, helping clinicians understand workload implications of model-guided care. For example, at 40% threshold, the model might identify 150 per 1000 high-risk patients

requiring intensive management while avoiding unnecessary referrals for 200 per 1000 low-risk patients, representing substantial healthcare resource optimization.

Confusion matrices at actionable thresholds should be reported at clinically relevant decision boundaries rather than overall accuracy. For example, the threshold distinguishing Stage 3a versus 3b (eGFR = 45 mL/min/1.73 m²) is critical for medication dosing adjustment decisions; certain antibiotics, anticoagulants, and chemotherapies require dose reduction at this threshold. Similarly, the threshold distinguishing Stage 4 versus 5 (eGFR = 15 mL/min/1.73 m²) triggers dialysis access creation and transplant evaluation. Clinical misclassification costs can be quantified as $\sum_{i,j} c_{ij} \cdot \text{confusion}[i, j]$, where c_{ij} represents the clinical cost of misclassifying stage i as stage j , with costs elicited from nephrologist expert panels. This aligns evaluation metrics directly with clinical consequences rather than treating all errors equally.

EHR integration and real-world deployment

Translating research models to clinical practice requires addressing implementation challenges beyond algorithmic performance. A pragmatic EHR integration pipeline includes four stages. First, pre-deployment external validation on the target population with site-specific recalibration if needed (e.g., temperature scaling fitted on local calibration set). Second, integration into clinical decision support (CDS) workflows, embedding the model into EHR systems to flag high-risk patients during encounters or automatically trigger nephrology referral orders for review. Third, guardrails and fail-safes including confidence thresholds (only display predictions with >80% maximum class probability), monotonicity audits to flag physiologically implausible predictions, and override mechanisms allowing clinicians to dismiss recommendations with documented rationale. Fourth, post-deployment monitoring to track model performance over time, detect dataset shift from changing patient demographics or laboratory calibration drift, and implement scheduled recalibration protocols (e.g., annually or when performance metrics degrade beyond pre-specified thresholds).

Regulatory considerations vary by jurisdiction. In the United States, the FDA classifies medical AI software as medical devices, and our model would likely fall under Class II (moderate risk) as it informs clinical decisions but does not automate treatment. Regulatory approval requires a pivotal validation study in the target population, demonstration of clinical utility (not just statistical performance), comprehensive risk-benefit analysis, and software documentation including intended use, contraindications, and limitations. Post-market surveillance for adverse events and algorithm change control protocols are mandatory. Similar requirements exist in the European Union (CE marking under Medical Device Regulation) and other jurisdictions.

Transparency and explainability are essential for clinical adoption. We recommend following TRIPOD-AI guidelines⁷⁴ for transparent reporting of clinical prediction models using AI/ML methods. A model card should document intended use population, training data characteristics, performance metrics stratified by subgroups, known limitations, and failure modes. Clinician training programs should educate end-users on when to trust versus override model predictions, how to interpret confidence estimates, and appropriate use cases versus out-of-scope scenarios. Implementation science research; qualitative interviews with clinicians, usability testing, workflow analysis; should inform deployment strategies to maximize adoption and minimize alert fatigue.

Broader implications for medical AI

This study's core principle; embedding structured clinical knowledge including stage ordering, renal physiology, and calibrated risk directly into model design; offers a template beyond CKD staging. The methodological framework is transferable to other ordinal medical classification tasks where misclassification costs scale with distance between categories.

In cardiovascular disease staging, NYHA functional class (I–IV) classifies heart failure severity, Canadian Cardiovascular Society angina grade (0–IV) classifies chest pain severity, and valvular stenosis grading (none/mild/moderate/severe) guides intervention timing. Predicting severe aortic stenosis when mild is present leads to inappropriate surgery; predicting mild stenosis when severe leads to delayed valve replacement and preventable heart failure. Ordinal neural networks with physiological constraints; for example, enforcing that higher N-terminal pro-BNP should not predict milder NYHA class; could improve reliability.

In cancer staging, the TNM system classifies tumor extent (T), lymph node involvement (N), and metastasis (M) as ordinal categories. Tumor grade (G1–G4 representing well-differentiated to poorly-differentiated histology) and treatment response assessment (complete response, partial response, stable disease, progressive disease) also have natural ordinal structure. Confusing complete response with progressive disease has catastrophic consequences (discontinuing effective therapy), whereas confusing partial response with complete response has modest consequences (continued monitoring instead of immediate surveillance). Ordinal-aware models could concentrate errors between adjacent categories, improving safety.

In neurodegenerative disease, Alzheimer's Clinical Dementia Rating scale (CDR 0–3) measures cognitive impairment severity, Parkinson's Hoehn and Yahr scale (1–5) stages motor progression, and ALS Functional Rating Scale tracks physical decline. Progression forecasting; predicting which Mild Cognitive Impairment patients will progress to Alzheimer's dementia within 3 years; is more clinically relevant than cross-sectional staging and avoids circular reasoning when biomarkers used for staging (e.g., amyloid-PET) are also predictive features.

The combination of ordinal structure to concentrate errors between adjacent categories, probability calibration to enable reliable risk communication, and physiological constraints to ensure clinical plausibility represents a principled approach to developing trustworthy medical AI that aligns with clinical reasoning rather than merely maximizing statistical performance metrics. However, the target leakage issue underscores a critical lesson: high statistical performance does not guarantee clinical utility or scientific validity. Our 0.904 Macro-F1 and 0.991 Macro-AUC represent excellent performance by conventional ML benchmarks, yet the model is

fundamentally limited by circular reasoning in its training data. This highlights the necessity of domain expertise in medical AI development; technical sophistication alone cannot overcome flawed problem formulations. Close collaboration between machine learning researchers and clinical domain experts from study design through validation is essential to ensure that models address meaningful clinical questions rather than optimizing statistically convenient but clinically circular tasks.

Conclusion

We present a framework for multi-stage chronic kidney disease (CKD) staging that unites ordinal classification, probability calibration, and physiological monotonicity within a knowledge distillation paradigm. By leveraging a calibrated CatBoost teacher to transfer temperature-scaled probabilities to an ordinal neural network student with an auxiliary eGFR regression head, our approach delivers accurate, well-calibrated, and physiologically plausible stage predictions on a cohort of 750 patients from Al-Ramadi Teaching Hospital. Across multiple performance metrics, the framework consistently outperformed strong baselines while retaining computational efficiency suitable for integration into clinical workflows. The integration of domain knowledge with modern learning techniques offers a template for developing safe, transparent, and effective clinical decision support tools. Broader validation across diverse populations and healthcare systems, and expansion to include additional KDIGO-recommended markers such as albuminuria, will be essential next steps toward real-world adoption. Our findings suggest that principled model design, grounded in both clinical and methodological rigor, can advance the reliability and trustworthiness of artificial intelligence in medicine.

Temperature <i>T</i>	Calibration Metrics			Discrimination		Confidence		Interpretation
	NLL	ECE (%)	Brier	Accuracy	MaxF1	Mean Conf.	Max Conf.	
0.5	0.891	7.23	0.0982	0.924	0.919	0.912	0.998	Severe under-smoothing: Probabilities remain too sharp, model is overconfident, poor calibration
0.8	0.702	5.84	0.0641	0.924	0.919	0.873	0.996	Under-smoothing: Still too confident, calibration improving but suboptimal
1.0	0.612	4.72	0.0521	0.924	0.919	0.841	0.994	No calibration (baseline): Raw CatBoost probabilities without temperature scaling
	[0.548, 0.681]	[3.8, 5.9]	[0.042, 0.064]					
1.5	0.512	3.82	0.0476	0.924	0.919	0.762	0.987	Improved: Better calibration, confidence more realistic
2.0	0.458	3.41	0.0458	0.924	0.919	0.694	0.978	Good calibration: Approaching optimal NLL and ECE
2.5	0.438	3.18	0.0449	0.924	0.919	0.641	0.967	Near-optimal: Very close to minimum NLL, excellent ECE
	[0.379, 0.502]	[2.3, 4.2]	[0.037, 0.054]					
3.0	0.426	3.08	0.0447	0.924	0.919	0.598	0.955	OPTIMAL (selected): Minimum NLL, excellent ECE and Brier
	[0.367, 0.489]	[2.1, 4.1]	[0.036, 0.053]					
3.5	0.431	3.27	0.0455	0.924	0.919	0.563	0.942	Slight over-smoothing: NLL increases slightly, still acceptable
4.0	0.445	3.58	0.0472	0.924	0.919	0.534	0.928	Over-smoothing: Probabilities becoming too uniform
	[0.384, 0.511]	[2.6, 4.7]	[0.039, 0.057]					
5.0	0.478	4.12	0.0513	0.924	0.919	0.489	0.905	Severe over-smoothing: Excessive smoothing degrades calibration
7.0	0.547	5.34	0.0604	0.924	0.919	0.421	0.871	Too smooth: Probabilities nearly uniform, poor calibration
10.0	0.631	6.89	0.0738	0.924	0.919	0.362	0.823	Extremely smooth: Model provides little discriminative information

Table 22. Comprehensive temperature calibration grid search on 15% Calibration Set ($N_{cal} = 79$). We optimized the temperature parameter T by minimizing multiclass negative log-likelihood (NLL) on a fixed calibration subset held out from the training data. For each candidate temperature, we computed NLL, expected calibration error (ECE) using 10 equal-width bins, Brier score, and maximum class probability (confidence). Temperature $T^* = 3$ achieves the minimum NLL while maintaining near-optimal ECE and Brier score. Values shown are point estimates; 95% confidence intervals computed via bootstrap (1000 iterations) are provided in brackets for key metrics.

Data availability

The patient-level data used in this study cannot be made publicly available due to ethical and privacy restrictions. Access may be granted upon reasonable request and subject to approval by the relevant institutional review board. Code availability: The code developed for this study has been deposited in a publicly accessible DOI-minting repository (Zenodo) and is available at: <https://zenodo.org/records/19247231>.

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Appendix

See Table 22.

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Author contributions

B.P.B. and O.L. conceived the study and designed the methodology. A.S.M. performed the data acquisition and preprocessing. B.P.B. implemented the models, conducted the experiments, and performed the statistical analysis. G.I. contributed to the formulation of the evaluation framework and interpretation of the results. A.R.-C. provided overall supervision, guidance on clinical relevance, and critical revisions to the manuscript. All authors discussed the results, contributed to the writing, and reviewed and approved the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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